

BEYOND THE BASE: EXPONENT ALLOCATION IN ROPE

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ABSTRACT

Redistributing RoPE’s interior frequencies improves extrapolation even when the sampled frequency range is fixed. We identify this allocation freedom and develop explicit constructions for learning and training-free deployment. Across three paired training seeds with identical sampled endpoints, changing only 30 interior frequencies lowers extrapolation loss at $2\times$, $4\times$, and $8\times$; matched-support interventions also improve task scores in frozen billion-parameter models. Full sine-cosine geometry reveals finite-window redundancy through an overlap-effective-rank identity and a shared slow-frequency limit. Cosh provides an analytic training-time allocation, reducing 16K perplexity from 138.8 to 95.6 in three-seed MLA training. For frozen deployment, we introduce TailSpline, the exact finite-grid solution of a one-sided boundary-smoothing objective. Without updating weights or fitting activations, it improves length-averaged RULER-13 over MrRoPE-Pro by 3.20 points on Llama-3-8B and 49.23 points on OLMo, with NIAH and PPL gains on both. Together, controlled identification, positional geometry, and constructive evaluations establish internal frequency allocation as a useful degree of freedom for RoPE design.

1 INTRODUCTION

Frequency redesign already enables longer contexts, through bandwise scaling, mixed-radix conversion and learned frequencies (Peng et al., 2024; Tian et al., 2026; Karypis et al., 2026). These methods often change range and internal placement together. We isolate the value of the interior allocation, then exploit it constructively: better learned extrapolation and training-free improvements in frozen models.

With initialization, data, optimization and actual frequency endpoints fixed, moving only 30 interior frequencies improves $2\times$, $4\times$ and $8\times$ extrapolation loss in all three seeds. Matched-support interventions also improve frozen OLMo and Qwen task scores. These gains identify value beyond range expansion; complete methods can then combine allocation with range, band and amplitude choices.

Finite-window positional directions can overlap despite uniform log-frequency spacing. This motivates two design routes: Cosh supplies a nonuniform basis during learning; TailSpline reshapes the transition into a frozen model’s fully interpolated low-frequency tail. Both retain the rotary operator and construct their tables without fitted model statistics.

The resulting benefits reach beyond a small controlled experiment. Cosh improves multi-seed MLA extrapolation and 750M continued training. TailSpline improves frozen Llama-3-8B and OLMo on RULER-13 and NIAH, while the symmetric boundary-matched construction (BM) raises frozen OLMo’s five-task natural-QA F1 from 21.62 to 25.44. Our contributions are:

- **Controlled allocation value.** Fixed-support training and frozen-model interventions isolate useful interior-frequency changes beyond range adjustments.
- **Structure and constructions.** Full-pair finite-window geometry characterizes positional redundancy; explicit density and boundary objectives yield Cosh and TailSpline allocations.
- **Learning and training-free gains.** Matched evaluations demonstrate improvements in language modeling, retrieval and natural QA, with complete length responses and protocol-specific controls.

2 ALLOCATION AS A DESIGN VARIABLE

For $K = d_{\text{rot}}/2$ pairs, standard RoPE uses $\omega_k = b^{-k/K}$. Any strictly ordered positive table admits

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (1)$$

The endpoints (a, R) fix the sampled log-frequency range; $K-2$ interior coordinates define allocation. Every geometric table has $z_k = k/(K-1)$ irrespective of base. The full space $\mathcal{Z}_K = \{0 = z_0 < \dots < z_{K-1} = 1\}$ includes analytic subfamilies and positive-gap learned parameterizations.

For deployment relative to a native table, write

$$\omega'_k = \omega_k^N s^{-m_k}, \quad d_k = \log(\omega_k^N / \omega'_k) = m_k \log s. \quad (2)$$

Changing frequencies preserves RoPE’s relative-position composition and norm preservation (Su et al., 2024). Multiplying cosine/sine amplitudes by g separately scales the rotary Q/K contribution by g^2 . Fixed-support experiments isolate z ; complete methods may jointly choose allocation, range and amplitude.

Geo denotes each protocol’s geometric reference and Native a released model’s inherited table. Standard and midpoint geometric grids share z but differ by an overall frequency offset. Exact grid and endpoint conventions are in Appendix D.2.

3 CONTROLLED EXTRAPOLATION AND FROZEN-MODEL GAINS

We first hold the frequency range fixed and change its interior allocation. The training comparison asks whether allocation improves learned extrapolation; the frozen comparison asks whether it improves tasks without updating weights.

3.1 A PAIRED INTERVENTION AT FIXED SUPPORT

We train paired 151.9M models with $K = 32$ rotary pairs at length 256. Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The geometric arm uses the FMRoPE training-base rule $b = L_{\text{train}}$ (Oka et al., 2026b); the other anchors Cosh quantiles with $\tau = 4$ to the same endpoints (construction in §5.1). Only the 30 interior exponents differ.

With the training table retained at evaluation, mean Cosh-minus-geometric tail NLL is +0.026 at the training length and $-0.281 / -0.176 / -0.146$ at 512/1024/2048 tokens. All three training seeds improve at every extended length (Fig. 1b; full contrasts in Appendix B.1). Thus interior allocation improves extrapolation at a small training-window cost, without changing either frequency endpoint.

3.2 THE EFFECT REMAINS IN FROZEN MODELS

On OLMo-2-1B and Qwen-2.5-1.5B, we fix the checkpoint, sampled endpoints, 64 pairs, attention amplitude, input rows, and decoder. The range extends from the native fast frequency to one-quarter of the native slow frequency. On OLMo’s nine held-out RULER tasks (Hsieh et al., 2024), uniform normalized exponents score 0.56%, a coarse ramp 61.04%, and a residual-guided profile 60.47%. Qwen’s four-task development comparison gives 57.75/64.00/66.50% (Appendix F.2). The simple ramp already captures much of the difference: the finding is the consequence of interior placement at fixed support.

Beyond one shape. The matched-endpoint M4 factorial tests two bases, two training lengths, three head dimensions and three paired seeds. Reference Cosh, a preassigned $1.25\times$ strength and matched exponential improve weighted OOD NLL in 7/12, 10/12 and 9/12 configurations. These short, 128-step runs support allocation breadth (Appendix B.3); the longer paired training above establishes the replicated extrapolation result.

A separate three-seed native Std-RoPE shape comparison also improves all five reported 512–8K extrapolation lengths, so the effect is not restricted to a midpoint geometric reference (Appendix C.1).

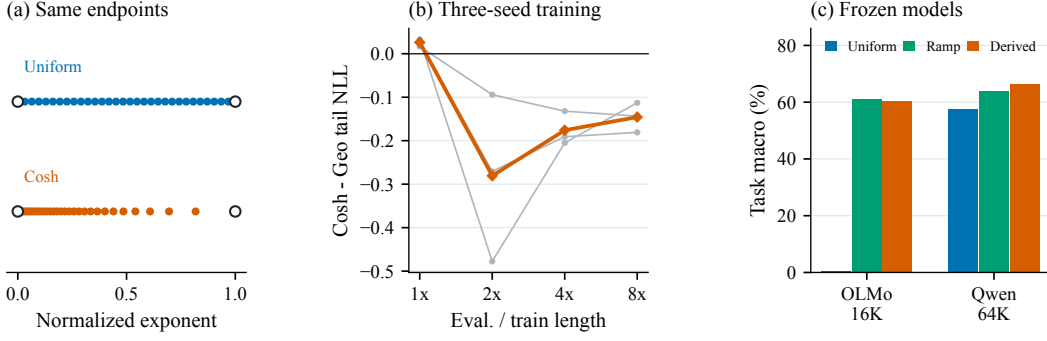


Figure 1: **Internal allocation improves extrapolation and frozen-model tasks.** (a) Uniform and anchored Cosh allocations share both sampled endpoints and 32 rotary pairs. (b) Cosh-minus-Geo tail NLL at the training length ($1\times = 256$) and three extended lengths. Thin lines are the three paired training seeds; the thick line is their mean. (c) Frozen matched-support interventions: OLMo has nine held-out tasks and Qwen four development tasks, each with 20 inputs/task. The uniform, coarse-ramp and derived arms share endpoints, amplitude, weights and inputs. Training-window costs remain visible in (b).

4 FINITE-WINDOW STRUCTURE OF ALLOCATION

The preceding experiments move frequency samples without changing their range. We now identify what those samples supply: finite-window positional subspaces whose overlap depends on placement.

4.1 COMPLETE ROTARY PAIRS AND POSITIONAL OVERLAP

For a fixed content pair, one rotary channel contributes $C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to an attention logit, where Δ is relative token separation. Both coefficients depend on content; its positional object is the full function space

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

Comparing only cosine curves would privilege one content phase. Instead, let $x_\omega(\Delta) = [\cos(\omega\Delta), \sin(\omega\Delta)]$ and define $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$, $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ under a declared separation measure. For nonsingular self-Grams, the singular values of $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ are the canonical correlations, or cosines of principal angles, of the two spaces (Björck & Golub, 1973). Their mean square,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

measures shared positional directions and is invariant to invertible basis changes within either pair. Let Γ be the Gram matrix of the concatenated block-whitened pairs, with diagonal blocks I_2 . The average pairwise overlap $\bar{c}$ gives the exact Rényi-2 effective-rank identity

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

Here $\text{tr}(\Gamma^2)$ is the frame potential of the normalized positional functions (Benedetto & Fickus, 2003). The identity specializes this classical measure to full rotary pairs: increasing pairwise overlap concentrates the same $2K$ coordinates into fewer effective positional directions.

4.2 FINITE-WINDOW CONCENTRATION OF SLOW DIRECTIONS

Proposition 1 (Shared slow-frequency subspace). *Under uniform measure on $[0, L]$, with $t = \Delta/L$, as $x = \omega L \rightarrow 0$, $\cos(xt) \rightarrow 1$ and $\sin(xt)/x \rightarrow t$, so V_ω converges to $\text{span}\{1, \Delta\}$. For $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$, squared chordal separation satisfies $2 - \|Q_{\omega\nu}\|_F^2 = \mathcal{O}(\epsilon^4)$.*

Distinct slow frequencies therefore approach a shared two-dimensional positional space. On the standard grid $b = 500,000$, $K = 64$, $L = 4096$, the 23 pairs with $\omega L \leq 1$ occupy 46 coordinates but have block-whitened rank 2.00013. The finite-window calculation also reaches the core experiment:

at $b = 256$, $K = 32$, $L = 256$, the eight slowest pairs span $\omega L \in [1.19, 4]$ and have rank 2.11 (Fig. 2a). Their overlap is substantial even though every frequency has $\omega L > 1$.

The geometry supplies a structural reason to reconsider log-uniform placement: distinct slow frequencies can provide strongly overlapping positional directions. It describes the supplied basis; learned content coefficients determine its use. Crossed tables and same-spectrum slot permutations demonstrate this compatibility dependence (Appendix I). We use the structure as a design motivation and validate the resulting allocations in models.

5 EXPLICIT ALLOCATION CONSTRUCTIONS

We construct tables in two settings: a model can learn a new allocation from training data, or a frozen model can receive a native-relative redistribution. Both change adjacent log-frequency intervals. The first designs a density and samples its quantiles; the second distributes a prescribed extension across a transition band.

5.1 AN ANALYTIC CONSTRUCTION FROM A TAIL-ENERGY PRIOR

The overlap analysis motivates spreading a finite budget across useful scales while controlling concentration toward the slow end. We express this motivation as a geometry-inspired, weight-independent density-design prior, using a density over normalized exponent space $[0, 1]$. Let $S_\rho(t) = \int_t^1 \rho(\phi) d\phi$, and consider the convex design criterion

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \int_0^1 S_\rho(t)^2 dt, \quad \text{s.t. } \int_0^1 \rho = 1, \rho \geq 0. \quad (6)$$

The first term penalizes concentration; the second penalizes cumulative slow-tail mass. Their balance selects how density shifts toward smaller exponents. The tail term is the quadratic form of the Dirichlet–Neumann Green kernel $\min(\phi, \psi)$ (Appendix A).

For $\alpha > 0$, $\beta \geq 0$, this criterion has a unique positive minimizer, $\rho_\tau(\phi) = \tau \cosh[\tau(1 - \phi)] / \sinh \tau$, where $\tau = \sqrt{\beta/\alpha}$. Its inverse CDF gives

$$Q_\tau(u) = 1 - \frac{\text{arcsinh}((1 - u) \sinh \tau)}{\tau}, \quad u_k = \frac{k + 1/2}{K}. \quad (7)$$

The uniform-density limit is recovered as $\tau \rightarrow 0$. Theorem 7 in Appendix A gives the complete construction proof.

EVQ-COSH converts the optimal density into a finite table through (7). We use Cosh for the analytic shape and state the grid and endpoint treatment for each finite EVQ-Cosh table. For $\tau > 0$, ρ_τ is decreasing, so its inverse CDF Q_τ is convex. For equally spaced nodes u_k , affine endpoint anchoring therefore gives

$$z_{\tau,k} = \frac{Q_\tau(u_k) - Q_\tau(u_0)}{Q_\tau(u_{K-1}) - Q_\tau(u_0)} \leq \frac{k}{K-1}, \quad \omega_{\tau,k} \geq \omega_{\text{Geo},k} \quad (8)$$

at fixed (a, R) : convexity places each quantile below the chord between the extreme nodes. Interior pairs move toward faster frequencies while endpoints remain fixed (Fig. 1a). Once τ , grid, and endpoint anchoring are specified, installation evaluates the table once without changing the rotary operator or trainable parameter count.

The strength τ sets this trade-off. We use declared reference settings and compare neighboring strengths and a second analytic family in Appendix B.3; Appendix A.12 gives the operating rules and their empirical basis.

5.2 TAILSPLINE FOR TRAINING-FREE DEPLOYMENT

A frozen model already uses its native frequencies. We retain its high-frequency band and divide its low-frequency tail by s , while designing the intervening allocation. Following the 32/1-turn boundaries used by MrRoPE (Tian et al., 2026), let l be the last pair making more than 32 turns over

the native window and h the first making less than one. Put $n = h - l$ and $q = \text{clip}(k - l, 0, n)$. The transition accumulates increments $\epsilon_q = m_q - m_{q-1}$, with $m_0 = 0, m_n = 1$.

The increments have a direct spectral meaning. For a geometric native grid with gap $c = \log b/K$,

$$x'_{l+q} - x'_{l+q-1} = c + \epsilon_q \log s. \quad (9)$$

Their unit sum allocates exactly $\log s$ of extra span across the band. Beyond it, $m = 1$ is constant, so the extra gap returns to zero.

MrRoPE-Pro uses increasing increments, $m_q = q(q+1)/[n(n+1)]$. We instead smooth the transition into the fully interpolated tail, whose subsequent increments are zero:

$$\min_{\epsilon: \sum_{q=1}^n \epsilon_q = 1} \sum_{q=1}^{n-1} (\epsilon_{q+1} - \epsilon_q)^2 + \epsilon_n^2. \quad (10)$$

This one-sided objective penalizes internal changes and the terminal jump. It leaves the high-frequency entry unpenalized, an explicit boundary prior rather than a fitted model statistic.

Theorem 2 (Finite-grid TailSpline). *The unique minimizer of Eq. (10) has positive increments and cumulative profile*

$$\epsilon_q = \frac{3(n+q)(n-q+1)}{n(n+1)(2n+1)}, \quad m_q^{\text{TS}} = \frac{q(3n^2 + 3n + 1 - q^2)}{n(n+1)(2n+1)}. \quad (11)$$

The quadratic objective is strictly convex; solving its tridiagonal stationarity system gives Eq. (11) (Appendix H). Adding the symmetric entry penalty ϵ_1^2 yields BM, whose increments are proportional to $q(n-q+1)$ (Appendix G.3). TailSpline reduces the terminal gap jump relative to MrPro by the exact factor $3/(2n+1)$: 0.086 for Llama’s $n = 17$ and 0.081 for OLMo’s $n = 18$. Figure 3a includes the two outer zero-increment regions so both entry and tail jumps are visible. This is a construction guarantee, whose task consequences are tested below.

Install $\omega'_k = \omega_k^N s^{-m_k}$ before prefill and retain it across all layers and input lengths for the declared s . We use $g = 1 + 0.1 \ln s$, matched to the baseline. Neither construction reads weights, activations or calibration outputs; the boundary rule uses only the native grid and reference length. The objectives prove construction properties; the following experiments test task value.

6 LEARNING AND TRAINING-FREE EVALUATION

6.1 LEARNING AN ALLOCATION: CURVES, CONTINUATION AND READOUT

A 432M MLA-style model allocates 16 rotary pairs, trains for 500M tokens at 8K, and uses three seeds (DeepSeek-AI, 2024a;b). Cosh improves all five extended lengths from 16K through 32K in every seed (Fig. 2b). Its 16K advantage is already present at half and three-quarters of training (Table 1). The same wavelength-blend rule applied to both learned tables preserves the advantage. This is a matched composition test, with the exact historical operator stated in Appendix E.1.

A 750M pair continues from the same 2K Geo checkpoint for 500M tokens at 4K. Cosh reduces 16K PPL by 45.9% and raises 8K greedy answer-token exact match from 0/40 to 31/40, with a small 4K PPL cost. This single-seed continuation includes 3% passkey and 10% downstream mix (Appendix D.5).

Allocation can also support complete answers after matched adaptation. The OLMo pair in Table 1 undergoes the same routing and answer-plus-EOS continuation. It uses physical inputs of at most 4K with explicit target-range position IDs, then evaluates real contiguous prompts. The 8/16K gains therefore demonstrate successful length transfer after phase exposure; they are separate from our zero-training results. Appendix F.1 retains the complete lineage, supervision and native-window controls.

6.2 TAILSPLINE ON FROZEN LLAMA AND OLMo

We compare exact finite-grid TailSpline and MrRoPE-Pro on frozen Llama-3-8B-Instruct, using $s = 4$, native window 8192, band $[18, 35]$, and common gain 1.1386294361. Each method installs

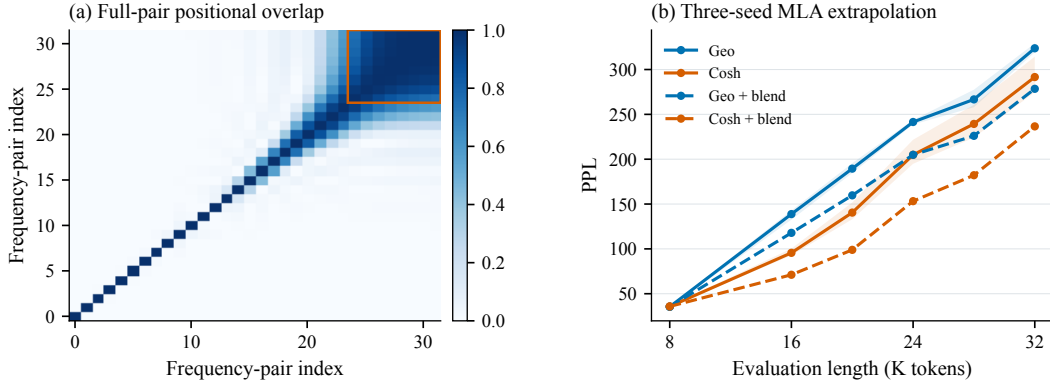


Figure 2: **Positional structure motivates allocation; model evaluations establish its value.** (a) Full-pair canonical overlap on the $b = 256$, $K = 32$, $L = 256$ identification grid under uniform separation measure. (b) Three-seed mean PPL of 432M MLA at 8–32K; solid curves use the trained tables, dashed curves the same wavelength blend ($s = 4$). Shading gives the three-seed range for unscaled tables, not a confidence interval. The panels use distinct protocols and do not identify overlap as a task-loss mediator.

Table 1: **Allocation benefits across learning protocols.** Each block compares matched arms, with its own metric. MLA uses three seeds, midpoint Cosh and 8K training. The 750M pair uses fixed endpoints and one seed. OLMo uses one matched adaptation pair with physical inputs ≤ 4 K and explicit long-phase exposure; its score requires the full answer and EOS. These blocks are not pooled.

Protocol	Endpoint	Geo / Native	Cosh
MLA, 50% budget	PPL at 16K	166.2	118.0
MLA, 75% budget	PPL at 16K	146.2	101.6
MLA, 100% budget	PPL at 16K	138.8	95.6
750M continuation	PPL at 4K	22.0	22.3
	PPL at 16K	45.1	24.4
	8K answer-token exact	0/40	31/40
OLMo matched adaptation	4K complete + EOS	95/100	100/100
	8K complete + EOS	18/100	98/100
	16K complete + EOS	0/100	60/100

one static table; weights, inputs, greedy decoder, precision and scorer are shared. RULER-13 has 10 paired prompts per task at 8/16/32K (390 per arm). Natural-text PPL uses 32 ProofPile-test and 14 PG19-test documents at the same lengths (138 rows per arm).

We aggregate lengths using normalized trapezoidal area in log length: weights $1/4, 1/2, 1/4$ at 8/16/32K. TailSpline raises RULER-13 AUC from 75.60 to 78.80%, a +3.20-point difference (paired 95% interval $[+0.65, +5.79]$). NIAH AUC improves by 3.75 points ($[+0.55, +7.03]$); its single-key passkey subtask has AUC 100.0 versus 97.5%. These are task-equal scores under the recorded scorers, including partial credit.

PPL AUC decreases from 4.62112 to 4.61664, with paired difference -0.00449 and interval $[-0.00872, -0.00121]$. This small gain is driven by 32K; 8/16K PPL is slightly worse (Fig. 3d). Full-13 improves at all three lengths. Across task families, retrieval, tracking and aggregation improve in AUC, while QA decreases by 1.25 points. The complete task and termination results are in Appendix H.

Prospective OLMo confirmation. The same frozen formula is tested on OLMo-2-1B-Instruct at $s = 4$, band $[14, 32]$ and 4/8/16K, with 390 RULER prompts and 138 LM rows per arm. The exact table was frozen before reading this OLMo evaluation; historical OLMo development had supplied direction priors. Full-13 AUC rises from 17.36 to 66.60% (+49.23 points, 95% inter-

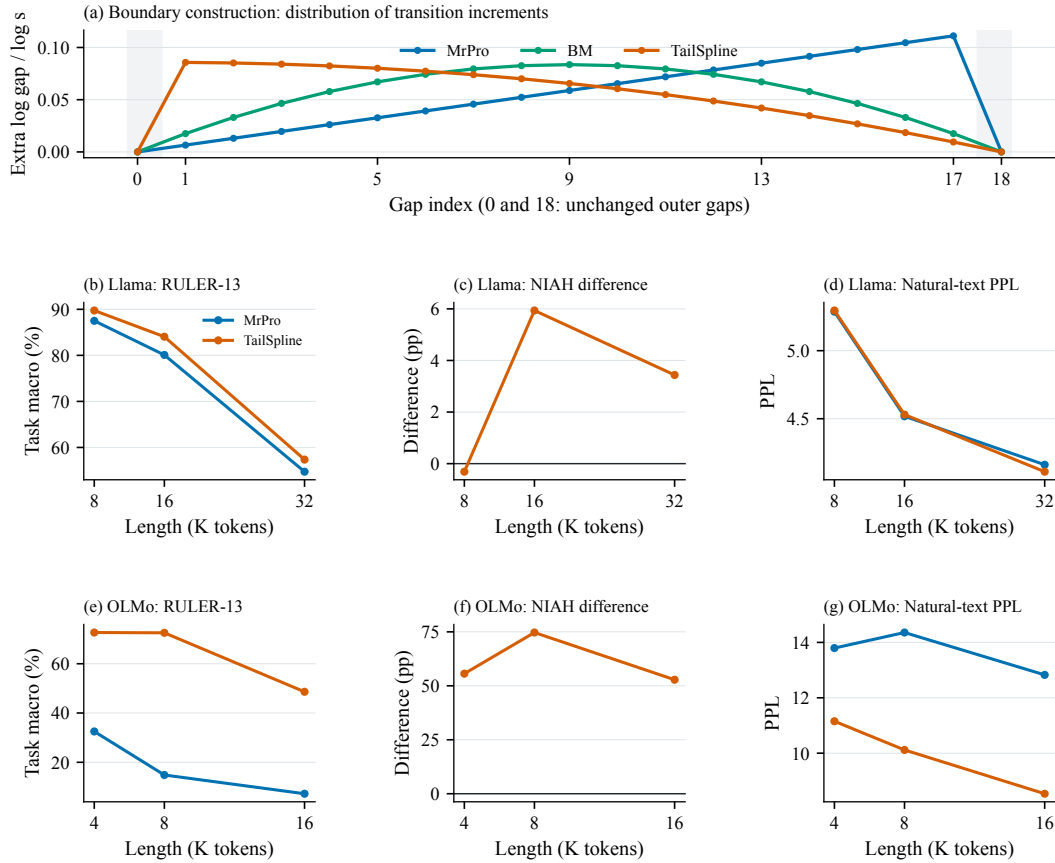


Figure 3: **Training-free gains on two model families.** (a) Analytic extra log gaps at $n = 17$, including unchanged outer gaps. The remaining rows show Llama (8/16/32K) and OLMo (4/8/16K). Columns show RULER-13 macro, NIAH paired difference, and PPL. Each comparison shares band, endpoints, gain and evaluation inputs; models have separate tokenized panels. NIAH is a RULER subset. Lines join measured points.

val [45.34, 53.04]). NIAH improves by 64.45 points and PPL AUC decreases by 3.853; all 13 task AUC differences are positive. This confirms the method direction on a second model family. TailSpline/MrPro reach the generation cap on 133/142 of 390 OLMo inputs, so official scores do not imply complete-answer-plus-EOS success. Full responses and interval definitions appear in Appendix H.

6.3 BM ON NATURAL QA AND BROADER EVIDENCE

On frozen OLMo, symmetric boundary matching improves each of five natural-QA task means (Table 2). The complete available untruncated pool contains 778 paired inputs; the reported long-input stratum contains 631 inputs exceeding 4K. With the same band, $s = 4$, amplitude and decoder, macro F1 rises from 21.62 to 25.44, a 3.82-point gain with paired interval [1.32, 6.29].

An equal-displacement control supplies a complementary shape test: BM and MrRoPE-Uni have the same endpoints, band, gain and total log-frequency displacement, yet score 51.32 versus 32.12% on the recorded six-task 16K panel (Appendix G.8). Thus total displacement alone does not explain that comparison.

These constructions have distinct tested scopes: BM’s natural-QA result remains a separate experiment from TailSpline’s two-model confirmation. Small Qwen 128K panels favor MrPro over BM (78.13/70.83% at 3B and 84.44/71.11% at 7B). Likewise, changing a table after learning can disrupt its association with Q/K coordinates; the range, crossed-table and slot controls in Appendix I explain

Table 2: **A concrete redistribution improves natural QA.** Frozen OLMo, static $s = 4$, whole-response token F1 (%). Both methods share the declared band, gain and inputs. The 631 inputs exceed the 4K native window; task means are equally weighted. Differences use unrounded scores. The paired, task-stratified 95% interval for the macro difference is [1.32, 6.29] points.

Task	n	MrRoPE-Pro	BM	Difference
HotpotQA	166	28.63	33.54	+4.91
2WikiMQA	173	21.55	25.43	+3.88
Qasper	119	13.94	19.06	+5.11
NarrativeQA	61	9.51	11.48	+1.96
MultiFieldQA	112	34.49	37.72	+3.23
Task-equal mean	631	21.62	25.44	+3.82

why training and frozen deployment need different constructions. They do not impose a fixed-support restriction on complete methods.

Additional evidence broadens the picture without mixing protocols. A frozen Qwen-0.5B configuration improves 64K RULER-13 over YaRN by 6.09 points (Appendix G.1); selective-Q/K adaptation improves long-input natural QA (Appendix F.1); Video DiT shows temporal extrapolation gains (Appendix D.7). M4 includes preliminary joint in-window/OOD improvements, while C42 isolates an internal-shape effect at fixed total displacement (Appendices B.3, G.7). These support the wider design space rather than a universal table or a mature in-window enhancement claim.

7 RELATED WORK

Frequency design and context extension. RoPE (Su et al., 2024) admits many frequency choices. Base-scaling analyses and FMRoPE connect frequency range, training length and extrapolation (Liu et al., 2024; Xu et al., 2024; Oka et al., 2026b). Position interpolation rescales all frequencies; NTK-aware and Dynamic NTK adjust the geometric base (Chen et al., 2023a; bloc97, 2023; emozilla, 2023). YaRN combines frequency-dependent interpolation with amplitude scaling (Peng et al., 2024). MrRoPE explicitly designs mixed-radix increments and compares uniform with progressive conversion (Tian et al., 2026). Nonuniform frequency design is therefore established. Our contribution is controlled evidence for gains beyond range changes, followed by explicit allocation objectives. TailSpline retains the standard outer-band policy but optimizes the transition’s low-frequency junction; it does not reintroduce the mixed-radix coordinate as a new idea.

Learned frequencies and positional structure. LeRoPE learns shared per-frequency scales and also trains with transferred fixed tables (Karypis et al., 2026); AdaRoPE studies head-specific frequencies and amplitude (Wang et al., 2026). These establish learned-frequency value, whereas our paired endpoints isolate interior allocation and our explicit constructors require no learned table. Studies of symbolic/positional attention and data-dependent bands examine how models use frequencies (Barbero et al., 2025; Urrutia et al., 2026; Oka et al., 2026a; Wu et al., 2026). DoPE uses entropy of model representations to guide denoising (Xiong et al., 2025). Our full-pair analysis concerns the supplied positional functions under a declared separation measure, using classical principal-angle and frame geometry (Björck & Golub, 1973; Benedetto & Fickus, 2003); compatibility experiments connect this basis to learned use.

Broader design spaces. LongRoPE and LongRoPE2 combine nonuniform scaling with search or continued training (Ding et al., 2024; Shang et al., 2025). FoPE, GRAPE, Selective RoPE and RePo alter the Fourier components, group action or position dependence (Hua et al., 2025; Zhang et al., 2026; Movahedi et al., 2026; Li et al., 2026). These are complementary to reallocating frequencies within the existing rotary operator. We compare complete methods under their actual learning or frozen-deployment protocols, rather than treating fixed-support identification as a restriction on subsequent design.

8 DISCUSSION AND CONCLUSION

Internal frequency allocation is useful: controlled experiments isolate its extrapolation gains, and explicit constructions realize benefits during learning and training-free deployment. Finite-window geometry identifies structure that allocation changes; Cosh, TailSpline and BM turn declared design objectives into executable tables. The empirical contribution includes both language modeling and actual task outputs.

The remaining opportunity is joint design. Range, transition boundaries, terminal scaling, amplitude and allocation can be chosen together, with full methods evaluated on native, intermediate and extended lengths. The present results establish external-length value; systematic mature in-window enhancement and cross-model selection of boundary priors remain future work. A complete task-optimality theory is not required to exploit the demonstrated freedom.

ETHICS STATEMENT

This work introduces no new dataset, deployed decision system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences. Improved long-context processing does not by itself establish factual reliability, privacy protection, or safe deployment; claims in this work concern the reported positional, language-modeling, and task endpoints.

REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. The accompanying source archive contains the active TeX, bibliography, plotted figures, numerical summaries, a SHA256 manifest, and standalone verification of the explicit Gram-matrix examples. Experimental checkpoints and public datasets are documented separately, together with the preprocessing required for selected reruns.

The core fixed-support identification uses six runs of 499,974,144 tokens each; the 432M comparison uses six 500M-token runs, and the 750M continuation uses two 500M-token runs. These are protocol-specific training budgets, with the corresponding recipes and additional studies detailed in the appendices.

AI USE STATEMENT

We used generative AI tools to assist with mathematical discussion, code development, literature organization, and manuscript editing. Experimental results are drawn from recorded model runs and numerical analyses. The authors are responsible for the correctness and originality of the submitted text, mathematical arguments, experimental claims, and accompanying code.

REFERENCES

- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3119–3137, 2024. doi: 10.18653/v1/2024.acl-long.172. URL <https://aclanthology.org/2024.acl-long.172/>.
- Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e6d58fc68c0f3c36ae6e0e64478a69c0-Abstract-Conference.html.
- John J. Benedetto and Matthew Fickus. Finite normalized tight frames. *Advances in Computational Mathematics*, 18:357–385, 2003. doi: 10.1023/A:1021323312367.
- Åke Björck and Gene H. Golub. Numerical methods for computing angles between linear subspaces. *Mathematics of Computation*, 27(123):579–594, 1973. doi: 10.1090/S0025-5718-1973-0348991-3.
- Sidney Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, Usvsn Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of BigScience Episode #5 – Workshop on Challenges & Perspectives in Creating Large Language Models*, pp. 95–136, 2022. doi: 10.18653/v1/2022.bigscience-1.9. URL <https://aclanthology.org/2022.bigscience-1.9/>.
- bloc97. NTK-aware scaled RoPE allows LLaMA models to have extended (8k+) context size without any fine-tuning and minimal perplexity degradation. Reddit, r/LocalLLaMA, 2023. URL https://www.reddit.com/r/LocalLLaMA/comments/141z7j5/ntkaware_scaled_rope_allows_llama_models_to_have/.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023a. doi: 10.48550/arXiv.2306.15595. URL <https://arxiv.org/abs/2306.15595>.
- Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. Longlora: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307*, 2023b. doi: 10.48550/arXiv.2309.12307. URL <https://arxiv.org/abs/2309.12307>.
- DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024a. doi: 10.48550/arXiv.2405.04434. URL <https://arxiv.org/abs/2405.04434>.
- DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024b.
- Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pp. 11091–11104, 2024. URL <https://proceedings.mlr.press/v235/ding24i.html>.
- mozilla. Dynamically scaled RoPE further increases performance of long context LLaMA with zero fine-tuning. Reddit, r/LocalLLaMA, 2023. URL https://www.reddit.com/r/LocalLLaMA/comments/14mrgpr/dynamically_scaled_rope_further_increases/.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. doi: 10.48550/arXiv.2407.21783. URL <https://arxiv.org/abs/2407.21783>.

- Robert M. Gray and David L. Neuhoff. Quantization. *IEEE Transactions on Information Theory*, 44(6):2325–2383, 1998. doi: 10.1109/18.720541.
- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pp. 6609–6625, 2020. doi: 10.18653/v1/2020.coling-main.580. URL <https://aclanthology.org/2020.coling-main.580/>.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesch, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=kIoBbc76Sy>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Ermo Hua, Che Jiang, Xingtai Lv, Kaiyan Zhang, Youbang Sun, Yuchen Fan, Xuekai Zhu, Biqing Qi, Ning Ding, and Bowen Zhou. Fourier position embedding: Enhancing attention’s periodic extension for length generalization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 24932–24949, 2025. URL <https://proceedings.mlr.press/v267/hua25b.html>.
- Petros Karypis, Sean O’Brien, Shreyas Kadekodi, Rui Zhu, and Julian McAuley. LeRoPE: Learnable RoPE frequencies improve language modeling. *arXiv preprint arXiv:2607.10134*, 2026. doi: 10.48550/arXiv.2607.10134. URL <https://arxiv.org/abs/2607.10134>.
- Huayang Li, Tianyu Zhao, Deng Cai, and Richard Sproat. RePo: Language models with context re-positioning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/64002>. arXiv:2512.14391.
- Xiaoran Liu, Hang Yan, Chenxin An, Xipeng Qiu, and Dahua Lin. Scaling laws of RoPE-based extrapolation. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://openreview.net/forum?id=JO7k0SJ5V6>. arXiv:2310.05209.
- Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017. URL <https://openreview.net/forum?id=Byj72udxe>. arXiv:1609.07843.
- Sajad Movahedi, Timur Carstensen, Arshia Afzal, Frank Hutter, Antonio Orvieto, and Volkan Cevher. Selective rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=AQo1SEElNb>.
- Yui Oka, Kentaro Hanafusa, Taku Hasegawa, Kyosuke Nishida, and Kuniko Saito. Probing rotary position embeddings through frequency entropy. In *International Conference on Learning Representations (ICLR)*, 2026a. URL <https://openreview.net/forum?id=1JZuEDq62N>.
- Yui Oka, Itsumi Saito, Kyosuke Nishida, and Kuniko Saito. Frequency bands in RoPE: Base frequency and context length shape the interpolation–extrapolation trade-off. In *International Conference on Learning Representations (ICLR)*, 2026b. URL <https://openreview.net/forum?id=PR1PPxvG9Q>.
- Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024. doi: 10.48550/arXiv.2406.17557. URL <https://arxiv.org/abs/2406.17557>.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=wHBfxhZulu>.

- Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen, and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 54203–54218, 2025. URL <https://proceedings.mlr.press/v267/shang25a.html>.
- Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pp. 843–852, 2015. URL <https://proceedings.mlr.press/v37/srivastava15.html>.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025. doi: 10.48550/arXiv.2501.00656. URL <https://arxiv.org/abs/2501.00656>.
- Qingyuan Tian, Wenhong Zhu, Xiaoran Liu, Xiaofeng Wang, and Rui Wang. MrRoPE: Mixed-radix rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/69413f87e5a34897cd010ca698097d0a-Abstract-Conference.html.
- Felipe Urrutia, Jorge Salas, Alexander Kozachinskiy, Cristian Buc Calderon, Hector Pasten, and Cristobal Rojas. Decoupling positional and symbolic attention behavior in transformers. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=V38yAooqddQ>.
- Shaowen Wang, Yuke Zheng, Tansheng Zhu, Shuang Chen, Shaofan Liu, Suncong Zheng, and Jian Li. AdaRoPE: Not all attention heads should rotate and scale equally. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL <https://icml.cc/virtual/2026/poster/60704>. arXiv:2607.19363.
- Suyuchen Wang, Ivan Kobzev, Peng Lu, Mehdi Rezagholizadeh, and Bang Liu. Resonance RoPE: Improving context length generalization of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 586–598, 2024. doi: 10.18653/v1/2024.findings-acl.32. URL <https://aclanthology.org/2024.findings-acl.32/>.
- Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Pan Zhang, Yuhang Cao, Jian Tong, Haodong Duan, Qipeng Guo, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. VideoRoPE: What makes for good video rotary position embedding? In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 66118–66136, 2025. URL <https://proceedings.mlr.press/v267/wei25h.html>.
- Xinyi Wu, Siyuan Liu, and Ali Jadbabaie. How data shapes RoPE frequency usage: From positional scale matching to length generalization. *arXiv preprint arXiv:2607.07678*, 2026. doi: 10.48550/arXiv.2607.07678. URL <https://arxiv.org/abs/2607.07678>.
- Jing Xiong, Liyang Fan, Hui Shen, Zunhai Su, Min Yang, Lingpeng Kong, and Ngai Wong. DoPE: Denoising rotary position embedding. *arXiv preprint arXiv:2511.09146*, 2025. doi: 10.48550/arXiv.2511.09146. URL <https://arxiv.org/abs/2511.09146>.
- Mingyu Xu, Xin Men, Bingning Wang, Qingyu Zhang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. Base of RoPE bounds context length. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2773. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/9f12dd32d552f3ad9eaa0e9dfec291be-Abstract-Conference.html.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. doi: 10.48550/arXiv.2412.15115. URL <https://arxiv.org/abs/2412.15115>.

Yifan Zhang, Zixiang Chen, Yifeng Liu, Zhen Qin, Huizhuo Yuan, Kangping Xu, Yang Yuan, Quanquan Gu, and Andrew Chi-Chih Yao. Group representational position encoding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/5cb58625f49ddf70fe2d527e9e4bbae5-Paper-Conference.pdf.

Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A free lunch for length extrapolation in video diffusion transformers. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, pp. 77539–77557, 2025. URL <https://proceedings.mlr.press/v267/zhao25m.html>.

GUIDE TO THE SUPPLEMENTARY MATERIAL

The supplementary material follows the paper’s argument, expanding each connection between geometry, learning, and practical allocation. Table 3 provides entry points to the complete results and proofs.

Table 3: **Where each part of the argument is developed.**

Question	Supporting material
How is allocation value identified?	Fixed-support training (App. B.1); multi-shape factorial (App. B.3); frozen controls (App. F.2).
How are the constructions derived?	Full-pair geometry and Cosh (App. A); TailSpline (App. H); BM (App. G.3).
What are the complete learning and deployment results?	Continuation and MLA (Apps. D, E.1); adaptation and readout (App. F); frozen profiles and QA (App. G); TailSpline (App. H).
What controls delimit the conclusions?	Range, crossed-table and slot interventions (App. I); profile diagnostics (App. J.6); interval design (App. J); video and architecture (Apps. D.7, D.9).

A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (12)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. Thus $c_{\omega\nu}$ is an intrinsic measure of overlap between the two positional subspaces.

Theorem 3 (Spectral budget identity). *For the block-whitened Gram matrix Γ with diagonal blocks I_2 , $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$. Hence its Rényi-2 effective rank is given by (5).*

Proof of Theorem 3. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 1. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (13)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (14)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 . Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (15)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (16)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (17)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1}M) + O(\epsilon^6) = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6), \quad (18)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \rightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{\overline{2\cos(\omega\Delta)}}{\omega^2} \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (19)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. The separation weights p therefore determine which polynomial directions survive centering. $\square$

Directional overlap and feature scale. For the original, unwhitened pair on $[0, L]$, put $x = \omega L$ and $\text{sinc } x = \sin x/x$. The self-Gram eigenvalues are

$$\lambda_\pm(S_\omega) = \frac{1 \pm |\text{sinc } x|}{2}, \quad \lambda_-(S_\omega) \sim \frac{x^2}{12}, \quad \kappa(S_\omega) \sim \frac{12}{x^2} \quad (x \rightarrow 0).$$

Thus the slow pair has a poorly scaled second direction before normalization. Block whitening removes this within-pair scaling and measures overlap between positional directions. Raw feature energy, coefficient magnitude, and numerical precision determine how strongly those directions can contribute to a model.

The standard $u_k = k/K$ grid gives 23 slow pairs in Figure 4; the endpoint-inclusive grid used in Figure 5 gives 24 at $K = 64$. Both figures use the uniform separation measure.

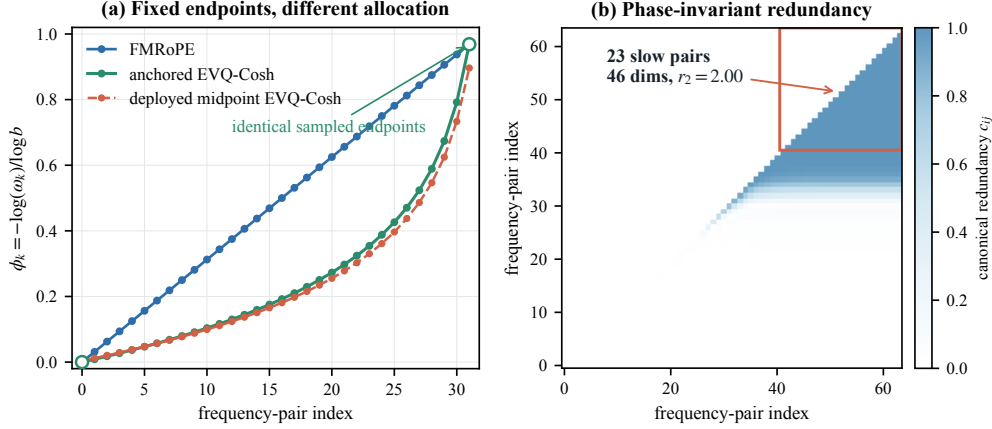


Figure 4: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

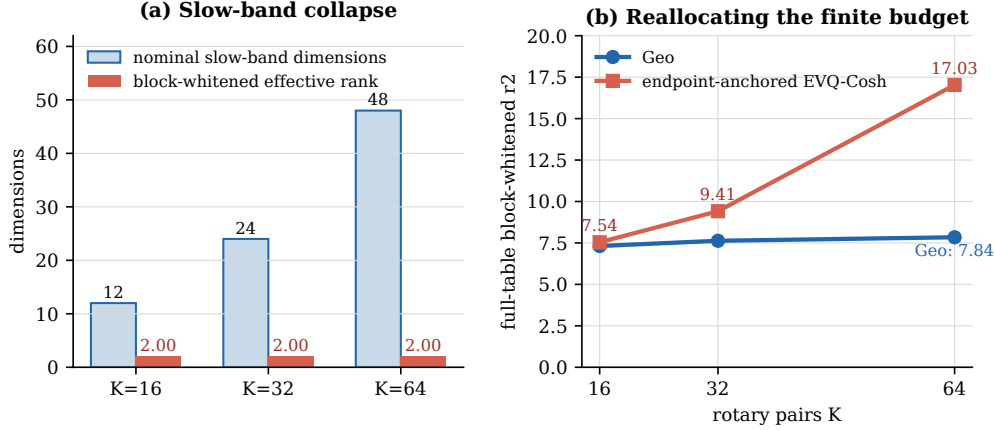


Figure 5: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows.

A.2 FINITE-WINDOW OVERLAP IN THE IDENTIFICATION EXPERIMENT

The slow-limit proposition has a finite-window counterpart that can be calculated directly on the identification grid. Set $b = 256$, $K = 32$, $L = 256$, and retain the n slowest frequencies of $\omega_j = b^{-j/K}$. Their dimensionless frequencies are $\omega_j L = 1.1892, 1.4142, 1.6818, 2, \dots$ in slow-to-fast order. The smallest is already greater than one, so we evaluate the full cross-Gram in (12) without a small-phase approximation.

For $n = 8$, the largest two Gram eigenvalues are 7.9681 and 7.5752, accounting for 97.1% of the trace; the third is 0.4246. As faster pairs enter, additional positional directions acquire mass. This is the finite-window overlap represented in Fig. 2. The accompanying `figs/make_story_figures.py` checks the $n = 8$ closed-form calculation against independent 256-node Gauss-Legendre quadrature and also recovers $r_2 = 2.00013$ for the 23 slow pairs of the $b = 500,000$, $K = 64$, $L = 4096$ grid. The machine-readable calculation is in `figs/finite_window_geometry.json`.

Table 4: **Finite-window positional concentration on the paired-training grid.** The separation measure is uniform on $[0, L]$. The nominal coordinate count is $2n$; r_2 is the effective rank of the block-whitened Gram.

Slowest pairs n	Largest ωL	Coordinates	r_2
2	1.41	4	2.00030
4	2.00	8	2.00361
6	2.83	12	2.02263
8	4.00	16	2.11474
12	8.00	24	2.89256
16	16.00	32	4.43339

A.3 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

Proposition 4 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (20)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (21)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (22)$$

so every entry of the cross-Gram in (12) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_\star((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (21) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k\Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 4 couples maximal rank on $[0, L]$ with recurrence beyond that interval. Its orthogonality also depends on the separation measure: weighting the same lattice by the triangular causal histogram introduces off-diagonal Gram terms. Thus interval coverage and pairwise rank capture different features of a positional basis.

A.4 TWO STATIC ORDERING COUNTEREXAMPLES

Two explicit examples show how phase coupling and interval length change the ordering of frequency tables.

Cosine-only collision can choose the lower-rank table. Take $K = 4$, $L = 8\pi$, and the two descending frequency tables

$$\Omega_A = (1, 5/8, 3/8, 1/4), \quad \Omega_B = (1, 6.01/8, 4.01/8, 1/4).$$

They share both endpoints. Define $C_{\cos}$ as the mean squared normalized cosine-cosine overlap over distinct pairs. For Ω_A , every cosine is an integer harmonic on $[0, L]$, so $C_{\cos}(A) = 0$ exactly.

Mixed harmonic parities nevertheless produce nonzero sine-cosine cross terms. For Ω_B , the small perturbation of an even-harmonic table gives

$$C_{\cos}(A) = 0 < 1.38725 \times 10^{-5} = C_{\cos}(B), \quad (23)$$

$$r_2(A) = 6.29478 < 7.99971 = r_2(B). \quad (24)$$

Thus a strict cosine-only preference can have the opposite full-subspace rank ordering. Both values follow directly from Eq. (12).

Full-subspace collision can reverse across lengths. Now use $K = 4$, $L = 2\pi$, and the fixed tables

$$\Omega_A = (1, 0.99, 0.02, 0.01), \quad \Omega_B = (1, 2/3, 1/3, 0.01).$$

Again the endpoints agree. Writing C_M for mean phase-invariant collision under the uniform measure on $[0, M]$, Eq. (12) gives

$$C_L(A) = 0.538996 < 0.631721 = C_L(B), \quad (25)$$

$$C_{2L}(A) = 0.384445 > 0.204995 = C_{2L}(B), \quad (26)$$

$$C_{4L}(A) = 0.345188 > 0.033506 = C_{4L}(B). \quad (27)$$

The close pairs in A retain high overlap as the interval grows, while the more evenly spread frequencies in B become easier to distinguish. The example makes the interval dependence of the geometry explicit.

All frequency entries and measures are specified above. The accompanying `figs/verify_explicit_geometry.py` evaluates the closed-form Gram and checks it against independent 128-point Gauss-Legendre quadrature; the largest discrepancy in the displayed metrics is below 4×10^{-14} . These finite basis examples complement the trained allocation comparisons and the weight-table crossings.

A.5 POST-HOC TRANSPLANT OBSTRUCTION

Table 5: Weights co-adapt to their training-time frequency table. Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head-query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic ($K = 32$, $b = 500,000$, $L = 512$). Static rank uses equal weights on integer separations $0, \dots, 511$, distinct from the continuous measure of the main geometric proposition. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

Theorem 5 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta)B = \mathcal{R}_\Omega(\Delta)$ on an open interval containing zero, then Ω' and Ω have the same frequency multiset up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Proof of Theorem 5. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta)A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace, including across its original two-dimensional blocks. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

Approximate compatibility within a geometric family. The exact kernel result leaves room for approximate recovery of a model’s behavior. In the historical 50M crossing, three geometric controls were derived from the Cosh table without fitting model loss: matching its slow endpoint ($b = 331.6\text{K}$), matching its log span (377.7K), and minimizing log-frequency RMS within the geometric family (8.06K). On Cosh-trained weights these give PPL 21.69, 22.14, and 9.63, compared with 23.05 for the original Geo table and 7.16 for the training Cosh table. A coarse geometric approximation can therefore recover substantial frozen compatibility. On Geo-trained weights the respective PPLs are 7.30, 7.16, and 45.63 (original Geo: 7.14). These historical tables also change support, so the fixed-support training and mature-model controls supply the separate interior-allocation estimates.

A.6 SLOT ASSIGNMENT AND CONDITIONAL PREDICTION ERROR

Let $\mathcal{R}(\Omega; M, D)$ denote the risk of a fixed model M on a fixed evaluation set D when frequencies Ω are assigned to labeled rotary slots. This domain includes unsorted assignments. It differs from the strictly ordered allocation domain in §2, where the full multiset recovers the sorted table.

Proposition 6 (Prediction error on an observed permutation pair). *Suppose a pair $\Omega, \pi\Omega$ has risk gap $\Delta_R = |\mathcal{R}(\Omega; M, D) - \mathcal{R}(\pi\Omega; M, D)| > 0$. For every permutation-invariant feature map Φ and every scalar predictor h ,*

$$\max_{\Xi \in \{\Omega, \pi\Omega\}} |h(\Phi(\Xi)) - \mathcal{R}(\Xi; M, D)| \geq \Delta_R/2.$$

Proof. Invariance gives the same prediction $c = h(\Phi(\Omega))$ for both assignments. The triangle inequality bounds Δ_R by the sum of the two prediction errors, and hence by twice their maximum. $\square$

The risk-gap premise is empirical. It does not follow from unequal projection norms or from Theorem 5. Synchronously transforming the Q/K blocks compensates the permutation exactly by (62); changing the table while freezing those projections measures a different intervention. In the measured pair, OLMo-1B PG-19 tail NLL changes from 3.10423 to 6.86493, and Qwen-2.5-1.5B 64K core-four macro changes from 0.7000 to 0.0000.

Discrete orbit-count sensitivity. In the archived scale-orbit construction at factor four, one-ULP float32 jitter changes the exact orbit count from 6 to 64, while the 16K tail-NLL difference (jitter minus exact) is -0.000013 (paired interval $[-0.003682, 0.003715]$) and the 16K core-four score remains zero. This rejects the interpretation that a large increase in this exact algebraic count necessarily produces an appreciable loss change on the tested finite horizon. It does not rule out a predictor that maps different counts to the same risk. Discrete-period design, as studied by Resonance RoPE (Wang et al., 2024), should be distinguished from this exact-ratio sensitivity test.

Finite coordinate perturbations and continuity. The clipped-affine C^2 approximation has movement MAE 0.001223 and RMSE 0.006118 relative to its reference profile. The evaluated OLMo retention score is 0.870971, below the 0.8750 floor. This finite example shows that this approximation tolerance did not certify retention. It establishes neither a Hessian eigenvector nor failure arbitrarily close to the reference. If fixed-input NLL is continuous and a reference has positive margin below an NLL gate, a sufficiently small neighborhood passes that gate by continuity.

A.7 FINITE-PHASE OPERATOR CHANGES

For $\delta = (\omega' - \omega)\Delta$, the second-order cosine remainder is bounded by $|\delta|^3/6$. The series converges for all δ , but a low-order approximation need not be accurate at large phase increments. An exact finite-phase operator bound is

$$\|\mathcal{R}_{\Omega'}(\Delta) - \mathcal{R}_{\Omega}(\Delta)\|_2 = 2 \max_j \left| \sin \frac{(\omega'_j - \omega_j)\Delta}{2} \right| \leq \max_j \min\{2, |\omega'_j - \omega_j| |\Delta|\}. \quad (28)$$

A.8 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

Theorem 7 (Unique density minimizer). *For $\alpha > 0$ and $\beta \geq 0$, the unit-mass nonnegative L^2 minimizer of Eq. (6) is uniquely $\rho_\tau(\phi) = \tau \cosh[\tau(1 - \phi)] / \sinh \tau$, with the uniform limit at $\tau = 0$.*

This subsection proves the theorem directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (29)$$

the allocation functional is (6), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (30)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_\phi^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (31)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (32)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (33)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (32) and (33) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (34)$$

which is main-text Theorem 7. Mass normalization enters the boundary condition through $g'(0) = \int_0^1 \rho = 1$. Positivity holds on the closed interval since $\rho_\tau(\phi) \geq \rho_\tau(1) = \tau / \sinh \tau > 0$ for $\tau > 0$ (with $\rho_\tau(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (34).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (35)$$

The kernel is the inverse of $-d^2/d\phi^2$ with Dirichlet condition $g(0) = 0$ and Neumann condition $g'(1) = 0$. Thus the tail term is an H^{-1} -type energy with these mixed boundary conditions: $\langle \rho, g \rangle = \int_0^1 |g'|^2 = \int_0^1 S_\rho^2$. This classifies the stated design prior without changing its empirical operating rule.

The strictly convex functional on $L^2([0, 1])$ therefore has the displayed stationary density as its unique constrained minimizer.

The Cosh solution has an analytically invertible CDF, so its quantiles turn the continuous allocation directly into a finite frequency table.

A.9 THE DENSITY OBJECTIVE IN INTERVAL COORDINATES

For a strictly positive density, let $Q = F^{-1}$ and $h(u) = Q'(u) > 0$ with $\int_0^1 h = 1$. Then $\rho(Q(u)) = 1/h(u)$ and $S_\rho(Q(u)) = 1 - u$. Changing variables in the declared density objective gives the exact identity

$$\mathcal{C}_{\text{app}} = \frac{1}{2} \int_0^1 \left[\frac{\alpha}{h(u)} + \beta(1-u)^2 h(u) \right] du.$$

Thus Cosh can equivalently be viewed as an interval-allocation design. Its quantile derivative is

$$h_\tau(u) = \frac{\sinh \tau}{\tau \sqrt{1 + (1-u)^2 \sinh^2 \tau}}.$$

This is a change of coordinates for the stated prior, not a derivation of that prior from exact full-pair overlap. TailSpline uses a different, discrete objective on increments of native-relative displacement. The common design object is frequency spacing; the two objectives serve distinct learning and deployment settings.

A.10 SINGLE-CROSSING BUDGET SHIFT

Lemma 8 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (36)$$

Proof of Lemma 8. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is strictly decreasing on $[0, 1]$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (36). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. $\square$

A.11 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 9 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (37)$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (38)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (39)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (37). $\square$

An immediate corollary follows.

Corollary 10 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (40)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2 \sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity provides a closed-form evaluation of the allocation energy.

A.12 ALLOCATION STRENGTH AND A REFERENCE RULE

The Cosh family varies with the strength τ . A reference used in the full-RoPE text experiments relates this strength to head width and training length:

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \frac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (41)$$

The reference sets $c = 1$ before training. Table 12 records the strength, grid, and setting procedure used in each experiment.

Local scaling calculation. The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (42)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (43)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1 - \phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (44)$$

The stiffness itself also has an exact expression:

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \arctan(\sinh \tau)}{\tau^2} - 1 \right]. \quad (45)$$

Indeed, unit mass reduces its defining integral to $\int \rho_\tau^{-1} - 1$, and integrating sech gives Eq. (45). The quartic term is a local expansion: at $\tau = \sqrt{2}, 2, 4$, it exceeds the exact stiffness by factors approximately 1.52, 1.97, 3.52. It is not a finite- τ accuracy guarantee for the operating rule.

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives the leading small- τ balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (46)$$

This calculation yields the scaling factor $d_{\text{head}}/\sqrt{L}$ with a local coefficient determined by $Q_1(L, b)$ and λ . The experiments evaluate the exact Cosh quantiles at the protocol’s chosen strength.

Reference and neighboring strengths. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at $0.75\times$, $1.25\times$, or $1.5\times$; the independent exact-range factorial

tests $0.75\times$, $1.00\times$, and $1.25\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entripoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. The reference and its neighboring strengths give a direct comparison of allocation intensity across the tested configurations.

A.13 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. We compare the measures in Wasserstein distance and approximate the density with the associated quantile-cell histogram ρ_K .

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (47)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (48)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (49)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (50)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoﬀ, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (51)$$

This expansion quantifies how finite exponent samples approximate a continuous allocation under a smooth distortion. For endpoint-anchored comparisons, $K = 2$ leaves no interior entries and the geometric and Cosh grids coincide.

Large- τ regime. At large τ , $m_\tau = \tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B CONTROLLED ALLOCATION EXPERIMENTS

B.1 EXACT-RANGE CONTROL AT 151.9M

Paired controls. The two trained tables have identical frequency endpoints. Within each seed pair the arms share: model architecture (151,898,880 parameters); trainable initialisation; training length

256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMROPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Writing Φ_τ for the inverse CDF in (7), let $q_k=\Phi_\tau((k + \frac{1}{2})/K)$, $s_k=(q_k - q_0)/(q_{K-1} - q_0)$, and $R=((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R\frac{k}{K-1}\right). \quad (52)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$. The anchors are common end positions selected with Python’s `random.Random(20260723)`: the eligible validation stream is divided into 32 equal-width bins, with one end position drawn per bin and a 2,048-token guard before the next bin. At anchor e , length L uses validation tokens $[e-L, e)$ and scores the final 128 targets $[e-128, e)$. Longer contexts therefore add earlier text around the same target tokens; the longest windows at distinct anchors are disjoint.

Exact range transformation.

$$R(L) = \frac{K-1}{K} \log L, \quad \omega_k^{\text{target}}(L) = \exp[-R(L)z_k].$$

Here $z_k = k/(K-1)$ for FMROPE and the anchored Cosh shape for its paired arm. The fixed condition uses $L = 256$ at every evaluation length; the target-matched condition substitutes the declared evaluation length. Both conditions use amplitude one.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Across all three seeds, fixed training range favors Cosh at every extended length, whereas target-matched range favors FMROPE. The paired design thus shows both an interior-allocation effect and a change in preferred allocation when the same trained weights are evaluated under a retargeted range.

Table 6: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

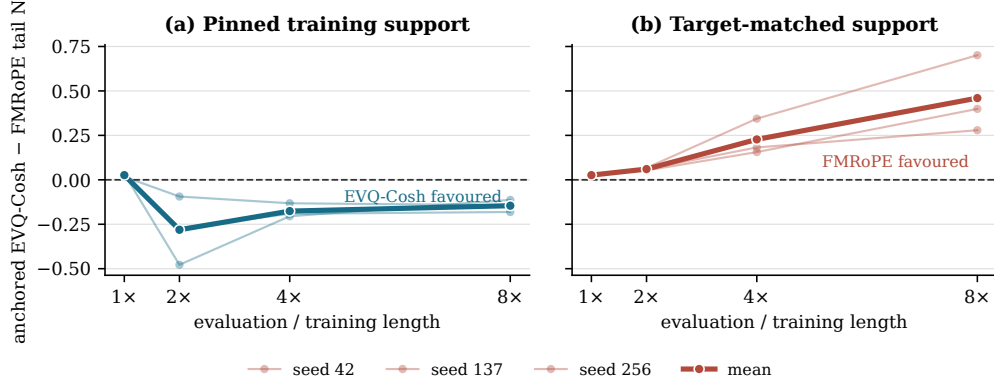


Figure 6: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

B.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMRoPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

For each training table Ω , the derived table uses the following label-free residual construction. Set $L_0 = 256$, $\Delta_j = 2j$ for $j = 0, \dots, 127$, and $p_j = (2L_0 - 4j - 1)/(L_0(L_0 + 1)/2)$, the grouped causal-separation weights. Let X_k have columns $\sqrt{p_j} \cos(\omega_k \Delta_j)$ and $\sqrt{p_j} \sin(\omega_k \Delta_j)$, and let X_{-k} collect all other pairs. Define

$$U_k = \frac{\|X_k - X_{-k} X_{-k}^\dagger X_k\|_F^2}{\|X_k\|_F^2}, \quad \bar{U}_k = \frac{U_k - \min_i U_i}{\max_i U_i - \min_i U_i}, \quad w_k = (1 - \bar{U}_k)^2.$$

The pseudoinverse uses relative singular-value cutoff 10^{-10} ; the zero-span convention is $\bar{U}_k = 0$. The frequency blend is $\omega'_k = \omega_k(1 - 3w_k/4)$, cast to FP32 with its endpoints set exactly to $\omega'_0 = \omega_0$ and $\omega'_{K-1} = \omega_{K-1}/4$. Both derived tables use $g = 1 + 0.1 \log 4$ and remain unchanged across the 512- and 1,024-token evaluations.

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. Both seeds exhibit the same preference for the runtime table derived from their training allocation.

Table 7: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	FMRoPE-derived table	Anchored-EVQ-COSH-derived table
FMRoPE-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

B.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at $0.75\times$, $1.00\times$ (the formula point), $1.25\times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Evaluation data and windows. Training and evaluation use the separate train and validation splits of cached wikitext-2-raw-v1, tokenized by GPT-NeoX-20B. Empty text rows are removed, remaining rows are stripped and concatenated without extra special tokens, and each split is repeated/truncated independently: 8,388,608 training tokens and 1,048,576 evaluation tokens. The data revision of that legacy Arrow cache is not recorded. For each arm, NumPy RandomState(9999) samples four distinct starts at each of $L, 2L, 4L, 8L$ in that order. Full next-token cross-entropy scores $4(rL - 1)$ targets at length rL , averaging the four chunk losses before exponentiation. All arms share these offsets. The split is distinct from training, but repeated validation text and potentially overlapping windows are not independent documents; uncertainty is computed over the structural configurations as specified below.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75\times$ 6.202858 (494.16); EVQ-COSH $1.00\times$ 6.202094 (493.78); EVQ-COSH $1.25\times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5\times$ arm is better; at formula $\tau=8$ the $0.75\times$ arm is. The improving direction points toward the intermediate strengths at both tested extremes.

The factorial varies architecture, base, and training length on a repeated WikiText-2 stream (Merity et al., 2017). Its configuration-level contrasts complement the longer FineWeb-Edu training comparison in §B.1.

The joint view retains native-window effects alongside OOD changes. Different configurations occupy distinct operating points, and some improve both quantities. It describes the measured grid rather than establishing a continuous frontier or a universal exchange rate between the two losses.

Table 8: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. B).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH 0.75 $\times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH 1.25 $\times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

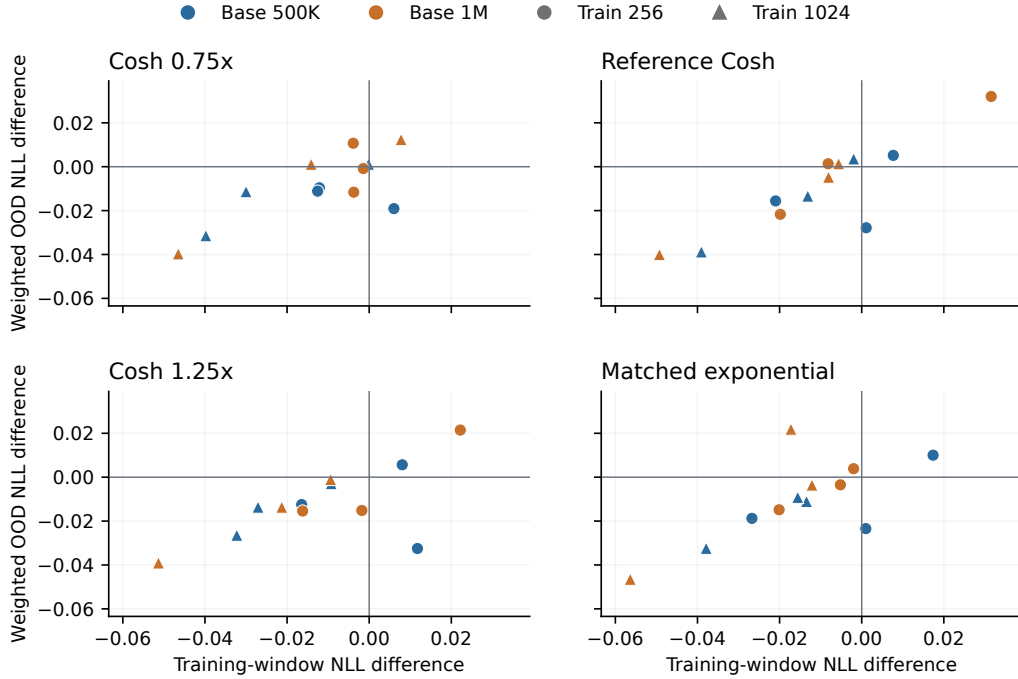


Figure 7: **Allocation changes training-window and extended-length loss jointly.** Each point is one of the twelve prespecified M4 configurations, averaging three paired seeds; all four nonuniform main arms are shown. Both axes are differences from the matching geometric arm, so negative values improve loss. OOD weights are proportional to $\log_2(r + 1)$ for $r \in \{2, 4, 8\}$, as in the original protocol. The horizontal coordinate is reconstructed from recorded four-decimal PPL; the vertical coordinate uses the recorded full-precision weighted NLL. Configurations are a designed grid, not independent draws from a model population; no pooled Pareto frontier or confidence ellipse is fitted.

B.4 FINITE DEPLOYMENT INTERVENTIONS

The permutation and finite- C^2 results in Appendix A.6 concern fixed-weight interventions. The C^2 approximation has its largest coordinate residual at slot 19. With zero-based indexing and native $K = 64$ geometric frequencies, that slot makes 13.25 turns in OLMo’s 4096-token window ($b = 500,000$) and 86.30 turns in Qwen’s 32768-token window ($b = 1,000,000$). Neither is the native one-turn boundary. A transition in an installed profile and the one-turn boundary of a native table are distinct quantities.

Three-band profiles retain a fast band, fully scale a slow band, and distribute the shift across an intermediate band. The cutoffs specify a construction; performance is measured with the checkpoint

and gain held explicit. On the OLMo six-task confirmation panel, BM improves 16K macro from 2.78% to 51.32% relative to MrRoPE-Pro. On the separate 631-prompt natural-QA long-input stratum it improves task-equal F1 from 21.62% to 25.44%. Detailed model contrasts and scoring protocols are in Appendix G; the exploratory twelve-profile panel is documented in Appendix J.6.

C LEARNED-FREQUENCY COMPARATOR

Table 9: Learned-frequency comparator (approximately 151.9M, $L_{\text{train}}=128$, FineWeb-Edu, 128 $\rightarrow$ 8K). Geo denotes the midpoint geometric grid. Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). The learned-inverse-frequency arm has one parameter per pair, shared across layers and heads.

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 ± 1.3	437.9 ± 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm learns one inverse frequency per rotary pair, shared across layers and heads. This is the learned-table parameterization also studied by Karypis et al. (2026). Its positional-parameter learning-rate sweep uses 10 $\times$ and 100 $\times$ the base rate; the stronger 100 $\times$ setting gives 455.3 PPL. The learned table and the fixed Cosh allocation both improve on Geo in this protocol.

These inverse frequencies are learned without fixing the sampled endpoints, so this comparison varies realized support as well as interior placement. The comparator uses 15M training tokens.

C.1 HISTORICAL NATIVE-GRID SHAPE CONTROL

A separate three-seed study compares the native standard-grid RoPE reference with a span-matched native-grid Cosh table. It uses 151,898,880 parameters, base 500,000, head dimension 64, training length 128 and 14,974,976 tokens. The same study compares additional analytic and kernel-derived shapes at matched span and RMS deformation. Selection uses 16 anchors and evaluation 32 disjoint test anchors. This is a short-training shape study, separate from the 500M-token fixed-support replication. The recorded seed-level 95% intervals for Cosh-minus-Std-RoPE NLL at

Table 10: Historical native-grid three-seed mean tail NLL. The complete length response retains the training window.

Table	128	256	512	1K	2K	4K	8K
Native Std-RoPE	5.5046	5.5026	5.6563	5.8456	6.0530	6.2566	6.3928
Native Cosh	5.4645	5.4473	5.5434	5.6966	5.8457	6.0664	6.2935

512/1K/2K/4K/8K are $[-0.200, -0.026]$, $[-0.183, -0.115]$, $[-0.241, -0.174]$, $[-0.260, -0.121]$, and $[-0.141, -0.058]$. All three seeds favor Cosh at each of those lengths. This supports the conclusion that the historical extrapolation improvement is not confined to a midpoint geometric comparator. It does not establish a universal Cosh optimum: the same report includes other useful nonuniform shapes. The source archive preserves the reported values and historical source identity in `figs/allocation_value_inputs.json`; the original training streams are separate.

D EXPERIMENTAL DETAILS

D.1 EXPERIMENT ORGANIZATION

This section gives the training, continuation, and video protocols. Apps. B and F detail the fixed-range and mature-model experiments.

D.2 TABLE AND OPERATOR IDENTITIES

RoPE, Geo and Native. Geo is a geometric RoPE table with $\omega_k = b^{-u_k}$. For standard $u_k = k/K$ and midpoint $u_k = (k + 1/2)/K$ grids at the same base,

$$\omega_k^{\text{mid}} = b^{-1/(2K)} \omega_k^{\text{standard}}.$$

Both are geometric RoPE with identical normalized allocation z and log-span R ; the midpoint offset changes a in Eq. (1). For $b > 1$, the factor is below one, giving uniformly slower rotation at the same token separation. The midpoint Cosh family’s $\tau \rightarrow 0$ limit is this midpoint Geo. Native identifies the inherited table of a specified released model. Each experiment retains its actual grid and baseline; geometric equivalence of the family does not by itself quantify a task-score difference.

Cosh shape and finite installation. The family in Eq. (7) uses $\phi_\tau(u) = 1 - \text{arcsinh}((1 - u) \sinh \tau) / \tau$. Grid choice and optional affine anchoring determine the installed tensor. At fixed endpoints, the anchored shape satisfies Eq. (8); a midpoint deployment without anchoring also changes the sampled range. Results are assigned to these actual installations, not to the family name alone.

Extension operators. Official-equation YaRN, the MLA wavelength blend in Appendix E.1, and the repository fixed-index ramp in Eq. (53) are distinct transformations. The last two appeared under historical YaRN labels in evaluation outputs; this manuscript names their actual operators. Paired gains with a shared operator remain evidence for that composition, and are not relabeled as official YaRN comparisons.

D.3 MODELS AND TRAINING SETTINGS

Table 11 collects the model settings and evaluation units. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 11: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime and intervention	Unit, endpoint, and appendix
Allocation identification	151.9M scratch; fixed support, 30 interiors	3 training seeds; tail NLL; App. B.1
Co-adaptation replication	151.9M frozen crossing; weights $\times$ runtime table	2 training seeds; tail NLL; App. B.2
Shape/configuration breadth	50.9M factorial; fixed support, 5 allocations	12 configs $\times$ 3 seeds; weighted OOD NLL; App. B.3
Scarce rotary budget	432M MLA-style; Geo/EVQ-COSH, $K=16$	3 training seeds; PPL; App. E.1
Full-parameter scale	750M/1.485B; continued/from-init Geo/EVQ-COSH	one comparison each; PPL/retrieval; Apps. D.5/F.1
Frozen allocation	OLMo/Qwen; same support and amplitude	20 rows/task; RULER; App. F.2
Zero-training deployment	OLMo; frozen long table, gain, session route	fixed checkpoint/rows; NLL/RULER/F1; App. F.2
Adaptation/source use	1.485B/8B; matched LoRA/source block	one trained pair; QA/RULER/NLL; Apps. F.1/F.3
Static index profile	Qwen-0.5B; one table across lengths	13 tasks, 20 rows/task/length; App. G.1
Profile placement	Qwen/Gemma; direct gap versus normalized index	4 tasks, 80 rows/task; App. G.2
Intermediate-band shape	OLMo/Qwen; matched BM/MrPro profiles	6 tasks; Table 20; App. G.4
Natural-QA transfer	OLMo; matched static BM/MrPro profiles	778 inputs, 631 long; token F1; App. G.5
Video modeling	129.6M video DiT; temporal Geo/EVQ-COSH	1 training seed; denoising MSE; App. D.7

Table 13 lists the training hyperparameters shared across the supporting text experiments.

Table 12: **Allocation protocols, parameter bounds, and float32 tensor SHA-256 digests. Top:** Cosh-family settings across reported experiments. Midpoint means $u_k = (k + 1/2)/K$; standard means $u_k = k/K$; anchoring fixes sampled endpoints after quantile evaluation. **Bottom:** Parameter bounds and float32 tensor hashes for static deployment profiles. MrRoPE-Pro parameters follow discrete turn cutoffs $\beta = 32, \alpha = 1$ with fast cutoff $l = \max\{j \mid \text{turns}_j > 32\}$, slow cutoff $h = \min\{j \mid \text{turns}_j < 1\}$, and transition width $n = h - l$. FullLagP2 evaluates all 32,768 integer lags, eliminating p2 aliasing spikes.

Part A: Cosh-Family Pre-training and Adaptation Settings			
Protocol	τ	Grid	Setting convention
151.9M fixed-range	4	Anchored midpoint	Fixed $64/\sqrt{256}$ reference
50.9M factorial	Varies	Anchored midpoint	Three preassigned multiples of $d_{\text{head}}/\sqrt{L_{\text{train}}}$
50M crossing	2.83	Midpoint	Reported training-table setting
432M MLA	1.414	Midpoint	Pre-specified architecture-specific setting
454M composition	1.5	Midpoint	Reported training recipe
750M continuation	1.5	Anchored inclusive	Fixed continuation recipe; $u_k = k/(K - 1)$
Llama-3-8B	1.414	Midpoint	Fixed adaptation recipe; approximately $128/\sqrt{8192}$
OLMo-2	2	Standard	Fixed $128/\sqrt{4096}$ reference
129.6M Video DiT	1.5	Midpoint	Best tested value in the video sweep
Part B: Static Deployment Profiles, Bounds, and Tensor SHA-256 Digests			
Profile / Architecture	Scale s	Bounds / Lags	Float32 Tensor SHA-256 Digest
MrRoPE-Pro (Qwen2.5)	4	$l=23, h=40, n=17$	33cbe3a4 0994ac2a 79126a14 ce302828 67bd6d49 b74c1ada 4d4cce8a 7e76016f ec9c5497 77f7a94c 73a1cc2d b5f22fef
MrRoPE-Pro (OLMo-2)	4	$l=14, h=32, n=18$	be202a7e 083e454f 2510034f 2cec5dfb ecd0c280 a11788e0 c4a869a3 d1628809 64ba3f37 1f589f7e 2f5ae471 d461096b
FullLagP2 (Qwen2.5)	4	32,768 integer lags	c80626f8 f9280854 577fd1e7 94439300 09a69370 65366adb 8eee5f62 ce7807a6 9229ec6a 6a7ce5c1 750273bc dbae9789 5a74a920 2722f14b 6ca2328e ce5ee98e
Boundary-Matched (OLMo-2)	4	$s=4, g=1.1386$	
Native RoPE (OLMo-2)	1	$L_0=4096, b=500K$	

Table 13: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.95, \epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 14 (750M: micro 7, accumulation 2), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Within the scratch and continuation comparisons specified in this section, the architecture, optimizer, data order, and training loop are fixed, and the intervention is the initialized inverse-frequency table. The released-checkpoint 1.485B comparison in Appendix F.1 uses separate trainer implementations, as detailed there. The scratch-text experiments use base $b=500K$ unless stated otherwise.

D.4 COMPOSITION WITH A FIXED-INDEX RANGE OPERATOR

The 454.2M decoder has 24 layers, width 1024, 16 heads, $d_{\text{head}} = 64$, and base $b = 500,000$. Geo ($\tau = 0$) and EVQ-COSH ($\tau = 1.5$) use midpoint frequency grids and train from initialization for 100M tokens per arm at length 2048, using FineWeb-Edu with a 10% synthetic-passkey mixture and seeds 42, 123, 7. The same inference-time operator is then applied to both trained tables.

For $k = 0, \dots, K - 1$, define

$$\ell = \lfloor 0.20K \rfloor, \quad h = \lfloor 0.90K \rfloor, \quad t_k = \text{clip}\left(\frac{k - \ell}{h - \ell}, 0, 1\right), \quad r_k = t_k^2(3 - 2t_k).$$

The repository’s fixed-index, YaRN-style smooth-ramp operator installs

$$\mathcal{R}_s(\omega_k) = \frac{\omega_k}{s^{r_k} T(s)^{r_k/2}}, \quad T(s) = 1 + 0.07 \log_2 s. \quad (53)$$

Here $K = 32$ and $s = 8$, giving $(\ell, h, T) = (6, 28, 1.21)$. The temperature factor is folded into the frequency divisor; cosine/sine amplitude stays at $g = 1$. Equation (53) specifies the operator used in this experiment, separately from the official-equation YaRN comparisons in §5.2. Table 14 reports the three-seed comparison.

Table 14: Matched fixed-ramp composition on the 454M three-seed experiment. $\mathcal{R}_8$ is defined in Eq. (53). PPL uses full-sequence scoring; PK is teacher-forced NLL-gap retrieval. Entries are three-seed means; PK@8K also gives the standard deviation.

Table	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41±5%	57%	51%	161.9	253.2
Geo+ $\mathcal{R}_8$	61±3%	59%	51%	82.9	157.7
EVQ-COSH	53±8%	63%	50%	150.3	229.5
EVQ-COSH+ $\mathcal{R}_8$	100±0%	79%	68%	70.9	107.5

D.5 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, fixed-length autoregressive answer-token exact match is 0% versus 77.5%. The EVQ-COSH continuation uses $\tau = 1.5$, $K = 32$, and the endpoint-inclusive quantiles $u_k = k/(K - 1)$. Writing q_τ for the inverse CDF in Eq. (7), the installed frequencies are

$$\omega_k = \exp\left[-\frac{K - 1}{K} \log b \, q_{1.5}\left(\frac{k}{K - 1}\right)\right], \quad b = 500,000.$$

This retains the Geo endpoints. Both arms are full-parameter continuations from the shared checkpoint. The autoregressive probe generates exactly the tokenized passkey plus its closing marker and compares every generated token ID with that expected sequence. It does not test a subsequent terminal EOS; the EOS-complete endpoint is the separate OLMo study in Table 25.

Table 15: 750M continued-pretraining check (seed 42; 2K→4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials per length. The pooled AR row combines 2K, 4K and 8K (120 trials: Geo 80/120, EVQ-COSH 111/120).

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Answer-token exact @8K	0%	77.5%
Pooled answer-token exact (2K/4K/8K)	66.67%	92.5%

D.6 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau = K_t / \sqrt{T} = 2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ (0.53 $\times$), which selects the supporting video configuration.

D.7 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames (4 $\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 16 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames under bidirectional 3D RoPE.

Table 16: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

D.8 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 5 separately measures incompatibility after a frozen post-hoc swap.

D.9 ALLOCATION UNDER GQA AND MLA COMPRESSION

A seed-42 125M study compares five attention configurations, each trained for 100M FineWeb-Edu tokens at length 2048 with 2% passkey mix. Within each configuration Geo and Cosh ($\tau = 1.414$) use the matched training recipe. Table 17 preserves the complete configuration comparison from the recorded experiment report.

Table 17: **Allocation effects across attention configurations.** Paired Geo/Cosh PPL at each length; the last column is the reported teacher-forced passkey retrieval aggregate, not autoregressive exact. One training seed per arm.

Configuration (K)	2K	4K	8K	16K	Retrieval (%)
MHA (32)	44.24/43.62	61.75/58.09	124.0/106.2	213.5/185.3	62.0/64.7
GQA-4 (32)	49.37/50.76	59.67/60.73	112.6/101.0	191.0/162.6	55.3/52.7
GQA-2 (32)	50.56/50.56	67.98/57.45	130.5/99.3	220.2/172.9	62.0/51.3
MLA-32 (16)	65.47/65.59	98.53/104.0	240.1/225.1	400.2/363.7	57.3/50.0
MLA-16 (8)	67.13/65.92	68.07/69.84	245.5/128.2	653.4/340.4	48.7/51.3

All five configurations improve long PPL, with the largest relative effect in the eight-pair MLA setting. The ordering across architectures is nonmonotone: KV sharing, latent compression, and rotary-pair count jointly define the setting. This supports studying allocation within architectural constraints, complementing the three-seed 432M experiment rather than establishing a one-variable scaling law for pair count.

E SUPPORTING RESULTS

E.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified $\tau = 1.414$ setting. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree.

Evaluation corpus and aggregation. The archived evaluator reads the shared cached FineWeb-Edu stream `val_fineweb-edu_5000000.pt`, nominally five million GPT-2 tokens. For each method and seed it resets NumPy RandomState to 9999, iterates through lengths 8192, 16384, 20480, 24576, 28672, and 32768, and draws eight distinct start offsets per length. Every chunk scores all $L - 1$ next-token targets, with full-sequence cross-entropy and no tail restriction. Its per-model PPL is the exponential of the mean chunk NLL; the table then takes the arithmetic mean and sample standard deviation of the three per-seed PPLs. Thus the requested targets per model are $8(L - 1)$ at each length; starts are sampled without replacement, but overlapping windows are possible. The same offsets are reused across methods.

The distributed preparation code tokenizes concatenated document text without added separators and prepares its evaluation stream from the FineWeb-Edu `sample-10BT` training split using shuffle seed 99999 and a 10,000-document buffer. This preparation rule does not certify document disjointness from training. The original cache’s upstream revision and token-array hash are not retained in the portable result snapshot; the reported experiment is therefore a shared-corpus length stress test. The recorded evaluator JSON is hash-bound, and its complete per-seed PPL values are provided in the source archive.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 18: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond, showing that the allocation also composes with this wavelength-dependent extension.

E.2 ADDITIONAL MODEL COMPARISONS

The following comparisons provide protocol-level detail for the main results.

Table 19: **Table and amplitude separated on the Qwen-2.5-3B 36-row panel (%)**. Matching the coefficient across tables makes the table contrast comparable; the last row varies amplitude alone at a fixed table.

Table	Coefficient c in $g = 1 + c \ln s$	32K	128K
MrRoPE-Pro	0.1	87.22	78.13
BM	0.1	91.67	70.83
MrRoPE-Pro	0.074	98.33	75.35
BM	0.074	100.00	70.00
BM	$g = 1$	89.58	58.82

Table 20: **Matched six-task RULER comparisons (%)**. Each column averages six tasks ($s = 4$; short/long prompt counts per arm).

Model	Lengths	Rows	Short		Long	
			MrPro	BM	MrPro	BM
OLMo-1B	4/16K	24/48	37.85	81.81	2.78	51.32
Qwen-3B	32/128K	12/24	87.22	91.67	78.13	70.83
Qwen-7B	32/128K	6/12	83.33	80.00	84.44	71.11

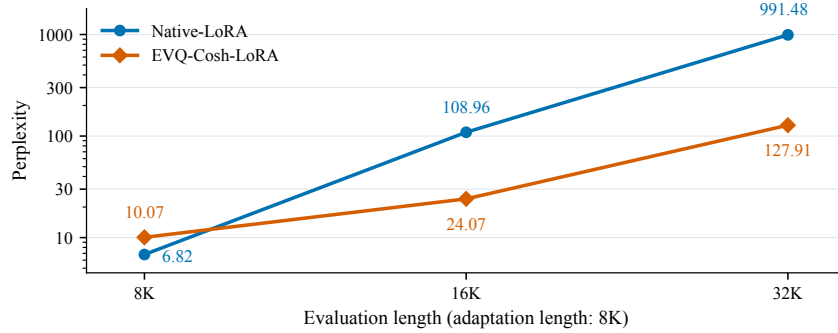


Figure 8: **Cosh allocation improves extrapolation under matched adaptation**. PPL is the exponential of equal-domain mean NLL.

Table 21: **Frozen-model RULER scores (%) at matched exponent range.** OLMo: held-out nine-task confirmation at 16K. Qwen: four-task development panel at 64K, 20 rows/task.

Exponent allocation	OLMo-1B	Qwen-1.5B
Geometric	0.56	57.75
Coarse ramp	61.04	64.00
Residual-guided	60.47	66.50

Table 22: **Static extension across native and extended windows.** Qwen-0.5B RULER-13 macro (%); g is fixed amplitude.

Installed table	g	32K	64K
Native	1.0000	54.78	22.05
YaRN-2	1.0693	55.94	45.37
Normalized-index-2	1.0513	55.92	51.46

Table 23: **Epistemic status of theoretical and empirical components.** Differentiating exact basis geometry, surrogate design criteria, heuristic operating rules, and downstream behavior.

Tier	Component	Object	Epistemic Status
1	Positional Basis	Subspace overlap $c_{\omega\nu}$, rank r_2 , limit V_0	Exact operator geometry (Prop. 1)
2	Surrogate Functional	Variational objective $\mathcal{C}_{\text{app}}[\rho]$ in (6)	Explicit design prior; not downstream $\mathcal{L}_{\text{CE}}$
3	Operating Rule	Cosh density ρ_τ and finite quantile grid	Unique density optimum; specified finite grid
4	Downstream LM	Tail NLL, RULER-13, natural LongBench QA	Measured under declared model and protocol

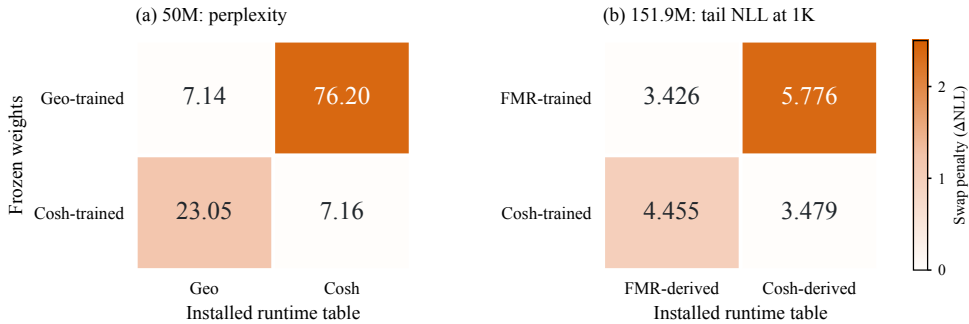


Figure 9: **Weights co-adapt to their exponent allocation.** Each row freezes one trained weight set and changes only its runtime table. Left: 50M crossing, PPL on eight windows. Right: 151.9M crossing, mean tail NLL on 32 anchors in each of two training seeds, using extensions derived from the two training tables. FMR is the geometric table at the fixed training range; shading shows the NLL increase relative to the matched diagonal cell.

F MATURE-SCALE PROTOCOLS

F.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the standard $u_k=k/K$ grid with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, with explicit position IDs providing target-range relative phases. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set.

Complete numeric answers after matched routing continuation. A separate follow-up to the routing parents applies the same 100-step query-gap continuation and 32-step answer-plus-EOS continuation to Native and EVQ-COSH (training seed 20260728). Every backward pass uses at most 4096 physical tokens; explicit position IDs expose target-range phases, with maximum observed position ID 16,257. Evaluation uses real contiguous prompts from the same numeric NIAH family and greedy decoding. The endpoint requires the complete expected answer string followed by observed terminal EOS, measured from generated token IDs.

Stage identities and supervision. The starting artifacts are the two 300-step counterfactual-routing parents (rank-64, alpha-128 Q/K/V/O LoRA). Their SHA-256 prefixes are `6570ab94aec6` (Native) and `95ceeb70117c` (Cosh); the later stages continue each arm’s own parent. Table 24 specifies the two matched follow-ups. Both use fused AdamW with $(\beta_1, \beta_2) = (0.9, 0.95)$, zero weight decay, gradient clipping 1, BF16, and cosine decay to 0.1 of peak LR after linear warmup. Global batch 8 uses microbatch 4 and accumulation 2. The trainable Q/K/V/O adapters contain 16,777,216 parameters; the base checkpoint is frozen.

Table 24: **The matched continuation producing complete-answer outcomes.** Training seed 20260728. Each cycle contains two counterfactual-routing updates and one natural-text update. Counts include both variants of a routing pair.

Stage	Steps	Peak LR / warmup	Supervision and position policy
Query-gap	100	$5 \times 10^{-5} / 20$	Answer CE plus counterfactual margin; semantic query-block offsets through 16K.
Answer + EOS	32	$10^{-5} / 4$	Answer CE, immediate EOS CE (weight 1), same counterfactual margin; realized source-to-answer gap curriculum.

Routing examples contain eight key-value blocks and one query. The two variants replace the queried source value while preserving the surrounding context; the answer changes accordingly. The training set has 1024 paired examples, the separate calibration set 128, with seeds 20293725 and 20304725 and different query templates. The final stage reuses the raw training rows with rebuilt answer-plus-EOS labels, covering three answer tokens and EOS; padding and other prompt tokens carry no answer loss. The counterfactual term uses margin 1 and weight 0.5, comparing the correct and swapped answer where their tokens differ. Natural replay uses ordinary next-token loss. The 100-step stage contains 67 routing/33 natural updates; the final stage contains 22/10.

Position-ID construction. Let q be the semantic query-block start, a the answer start, h the end-exclusive source span, and n the active sequence length. Context positions before q remain contiguous. Both counterfactual variants share the position array:

$$p_i = i + o \mathbf{1}\{i \geq q\}.$$

In the 100-step stage, each routing update uses four pair offsets: a low offset alternating between 0 and $[1, 4096]$, one from $[4097, 8192]$, and two from $[8193, 12289]$. Values traverse each band by a seed-derived coprime modular stride, with a deterministic per-update permutation. In the final stage, the stream instead specifies realized gap targets: the low target alternates between contiguous and $[4096, 8191]$, one target lies in $[8192, 12287]$, and two in $[12288, 16079]$. For a noncontiguous target g , set

$$o = \min\{g - (a - h), 12289, 16384 - \min(n, 4095)\}.$$

The implementation requires the resulting gap to remain in the target band and all active position IDs below 16384. Natural replay remains contiguous. The exact two deterministic schedules are included in `figs/verify_routing_schedule.py`; recomputation matches the run-record hashes for all 268 query offsets and all 88 final-stage gap targets. This procedure exposes target-range phases during training while keeping physical sequences at most 4096 tokens.

The evaluation uses separately generated, real contiguous numeric-NIAH prompts, 100 per length, with greedy decoding and complete-string-plus-EOS scoring. Training uses custom counterfactual query templates rather than the official evaluation template. The source receipts and full parent/final adapter hashes are included in `figs/routing_protocol_receipts.json` and `figs/recovered_asset_inputs.json`.

Table 25: Complete-answer-string plus terminal-EOS exact match after the matched routing follow-up, 100 inputs per length and one trained pair.

Length	Native	EVQ-COSH
4K	95/100	100/100
8K	18/100	98/100
16K	0/100	60/100

All final generations in both arms terminate with EOS. On the EVQ-COSH subsets whose source-query gap exceeds the original training range, exact match is 48/50 at 8K and 31/66 at 16K. These outputs belong to the additional continuation above, separately from the earlier first-number retrieval comparison.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled `Q/K metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 26 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals quantify evaluation-row variation for the trained pair.

Table 26: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on. The Llama rows use the separate RULER-family continuation: 516 steps, rank-64 Q/K/V/O LoRA, length 8192, seed 20420726. The 300-step LongAlpaca adaptation supplies the temporal-PPL and gold-block results, not these RULER scores.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is +0.0724/+0.0381/−0.0437/−0.1351 at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K.

The EVQ-COSH branch uses a Hugging Face single-GPU loop; the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. This comparison includes the frequency-table change and the two trainer implementations.

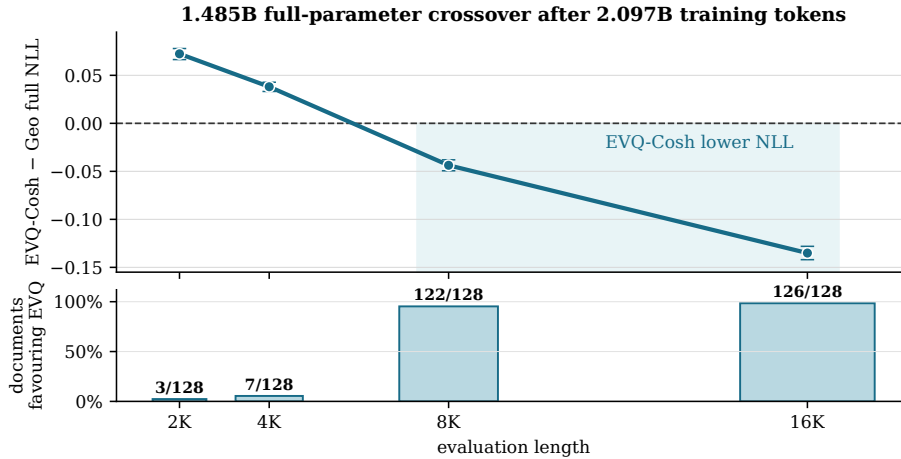


Figure 10: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

F.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 21 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The reference rows use the original Native table and the Transformers YaRN frequency

map. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $w_k = (1 - \tilde{U}_k)^2$ in the frequency blend of Eq. (54), so less unique pairs move farther.

Numerical construction of the fixed-support controls. The following specifies the derived arm independently of the later FullLagP2 construction. Use the checkpoint’s actual float32 native tensor promoted to float64, with $L = 4096$ for OLMo and 32768 for Qwen. Start from integer separations $d = 0, \dots, L - 1$ and weights $L - d$. Group consecutive separations in blocks of two, retain each block’s rounded weighted-centroid separation, and sum its weights; normalize the weights to unit mass. This gives 2048 and 16384 weighted support points, respectively.

Build X with columns $\sqrt{w_d} \cos(d\omega_k)$ and $\sqrt{w_d} \sin(d\omega_k)$ in float64. For each two-column block X_k , solve $B_k = \arg \min_B \|X_k - X_{-k}B\|_F$ with singular-value cutoff `rcond` = 10^{-10} , and compute

$$U_k = \text{clip} \left(\frac{\|X_k - X_{-k}B_k\|_F^2}{\|X_k\|_F^2}, 0, 1 \right).$$

If the marginal energy is zero, set $U_k = 0$. Min–max normalize the U_k values; if their span is zero, use the zero vector. Then $w_k = (1 - \tilde{U}_k)^2$ and $\omega'_k = \omega_k(1 - w_k) + \omega_k w_k/4$. Cast to float32, overwrite the first frequency with the exact native first entry and the last with the float32 native last entry divided by four, and require strict positivity and decreasing order. The resulting tensor SHA-256 prefixes are `a435d7544144` (OLMo) and `3512335e408c` (Qwen).

The uniform control is log-linear between the same endpoints. The coarse control enumerates all integer ramp bounds $0 \leq l < h < K$, minimizes mean squared distance to w_k in movement space, and breaks ties lexicographically in (l, h) . It uses the same frequency blend and endpoint overwrite. The selected bounds are (20, 22) for OLMo and (28, 31) for Qwen. FullLagP2 in Appendix G.9 instead uses every integer separation and installs a log-frequency shift; its measurements are a separate experiment.

Table 27: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Table 21. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalued, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are [54.88, 64.80] points for OLMo and [0.25, 17.50] points for Qwen; the paired derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 28 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 28: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The OLMo deployment policy selects the exact Native table when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Table 29: Compact zero-training deployment endpoints on the frozen OLMo checkpoint, using the specified table, amplitude, and session routing.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$. The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Table 21 gives the corresponding allocation-only comparison.

F.3 LLaMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. Native uses the pretrained endpoint grid, and Cosh uses midpoint quantiles; the same adaptation recipe is applied to these two frequency tables. The frozen adaptation tensor is drawn from LongAlpaca-12k (Chen et al., 2023b): 7,476 training and 152 validation rows after length filtering, with effective batch 8 (microbatch 2, accumulation 4).

Temporal holdout and aggregation. Evaluation uses arXiv titles/abstracts, Federal Register titles/abstracts, and Stack Overflow question bodies selected by publication or creation timestamps between January 1 and July 12, 2026. These are separate from the LongAlpaca adaptation corpus. Collection was frozen before evaluation; each domain has eight disjoint-document 32,768-token packs formed by concatenation with EOS separators. Positions and causal attention continue across

document boundaries. The 8K and 16K conditions are exact prefixes of the same 32K packs. All ordinary text targets in each prefix are scored, excluding document-first tokens and inserted EOS separators. For domain d , let $\ell_d(L)$ and $n_d(L)$ be the summed token loss and valid-token count over its eight packs. The reported aggregate is

$$\text{NLL}(L) = \frac{1}{3} \sum_{d=1}^3 \frac{\ell_d(L)}{n_d(L)}, \quad \text{PPL}(L) = \exp(\text{NLL}(L)).$$

Table 30 gives token counts and the distribution of the 24 paired pack-level NLL differences. The accompanying `figs/llama_temporal_summary.json` retains each pack’s loss sum and count, its domain identity, and the frozen collection hash.

Table 30: Llama-3-8B temporal holdout: scored tokens and the 24 paired pack NLL differences (EVQ-COSH minus Native). Each length uses the same packs.

Prefix	Valid tokens/arm	Valid tokens/pack	Pack ΔNLL : min / median / max
8K	194,794	8,051–8,183	+0.193 / +0.432 / +0.613
16K	389,672	16,111–16,353	−2.069 / −1.436 / −1.278
32K	779,400	32,243–32,679	−2.319 / −2.074 / −1.754

The domain-macro EVQ-COSH-minus-Native NLL is +0.390 / −1.510 / −2.048 at 8/16/32K.

Position-preserving source intervention. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry. NLL pools the negative log probabilities of the 30 answer tokens (3 per case); deletion changes this mean by −0.0095 for Native and +1.5055 for EVQ-COSH. The EVQ-COSH change is positive in all ten cases. Case-level hit@16 ranges are 3.12–34.38% for Native and 62.50–68.75% for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

Generation and natural-QA outcomes of the same adapters. The same three-arm source collection also includes the unadapted base model. Its domain-macro prefix PPL at 8/16/32K is 7.948/150.545/1492.915, compared with 6.817/108.958/991.475 for Native-LoRA and 10.068/24.068/127.911 for Cosh-LoRA. This adds the adaptation baseline while retaining the 8K cost; the matched Native-LoRA comparison remains the control for generic adaptation. The portable three-arm prefix summaries retain their source collection identity.

The source interventions reveal how remote evidence affects answer probabilities. Task evaluation measures whether those probabilities produce the required output. In the seed-42 capability evaluation, Cosh passkey exact is 100/0/0% at 8/16/32K, and single-needle top-1 is 56.67/3.33/0%. The registered three-arm QA comparison gives task-macro F1 0.2309 for the base model, 0.2110 for Native-LoRA, and 0.1126 for Cosh-LoRA; the paired Cosh-minus-Native difference is −0.0984, with 95% bootstrap interval [−0.1297, −0.0697]. The deficit is concentrated in inputs of at most 8K; on the longer subset all arms are near the floor. Forced-gold inclusion changes NLL by −0.034, and the sparse readout pilot does not improve conversion. These report-backed outcomes accompany the probability and source-use measurements of the same 300-step adapters. They motivate distinguishing access to a distant source from successful task readout.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 26 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

G FROZEN EXPONENT-ADJUSTMENT PROTOCOLS

This section specifies the static profiles and evaluation panels used in §5.2. An amplitude g multiplies both the cosine and sine tables, so its contribution to a rotated Q/K logit is g^2 . Every table is installed before prefill and retained for the entire request. The accompanying `figs/recorded_runtime_identities.json` records checkpoint, tokenizer, and evaluator identities. The index/placement studies use RULER commit `c3f5e3b4f87f97e048793bb510a3a6b19a46bf3a`.

For the native-relative displacement in (2), the three installation laws are

$$d_k = \begin{cases} -\log(1 - w_k + w_k/s), & \text{frequency blend,} \\ m_k \log s, & \text{log-frequency shift,} \\ \sum_{j < k} \log \lambda_j, & \text{mixed-radix conversion.} \end{cases} \quad (54)$$

The blend weight w_k and log-shift profile m_k have different meanings; the first line is the exact conversion between them.

G.1 NORMALIZED-INDEX PROFILE

For a native table with K pairs, base b , and reference length L_{ref} , the normalized-index construction evaluates the profile on a 64-pair reference grid with the same b, L_{ref} :

$$\bar{\omega}_j = b^{-j/64}, \quad c_{64} = (1 - b^{-1/64})^{-1}, \quad \xi_j = \log \frac{L_{\text{ref}} \bar{\omega}_j}{2\pi c_{64}}, \quad (55)$$

$$\bar{m}_j = \text{clip} \left(\frac{\xi_H - \xi_j}{\xi_H - \xi_L}, 0, 1 \right), \quad \xi_H = 0.7382780681078285, \quad \xi_L = 0.366403835112904. \quad (56)$$

These two constants were fitted to the existing OLMo displacement profile and then frozen. Linear interpolation of $(j/63, \bar{m}_j)$ at $k/(K-1)$ gives m_k on the target grid. The installed table is $\omega'_k = \omega_k^N s^{-m_k}$.

The Qwen-2.5-0.5B-Instruct experiment uses its native $K = 32$ pairs, $b = 10^6$, and original context length $L_{\text{ref}} = 32768$, with $s = 2$ and fixed amplitude $g = 1 + 0.074 \log 2 = 1.0512928914$. The YaRN-2 comparison uses the Transformers official-equation table and $g = 1 + 0.1 \log 2 = 1.0693147181$; Native uses $g = 1$.

The full RULER-13 confirmation uses seed 202609027, 20 examples per task and length, the same tokenizer and native chat template, and greedy task-budgeted generation. It contains 520 generations per arm across 32768 and 65536 tokens. Per-task scores are shown in Table 31. The macro averages the thirteen task means equally. Paired intervals use 10,000 bootstrap samples within task strata with seed 202609028. The 64K index-minus-YaRN difference is 6.0897 percentage points, with interval $[2.7627, 9.5835]$; at 32K it is -0.0256 points, with interval $[-3.2630, 3.2179]$.

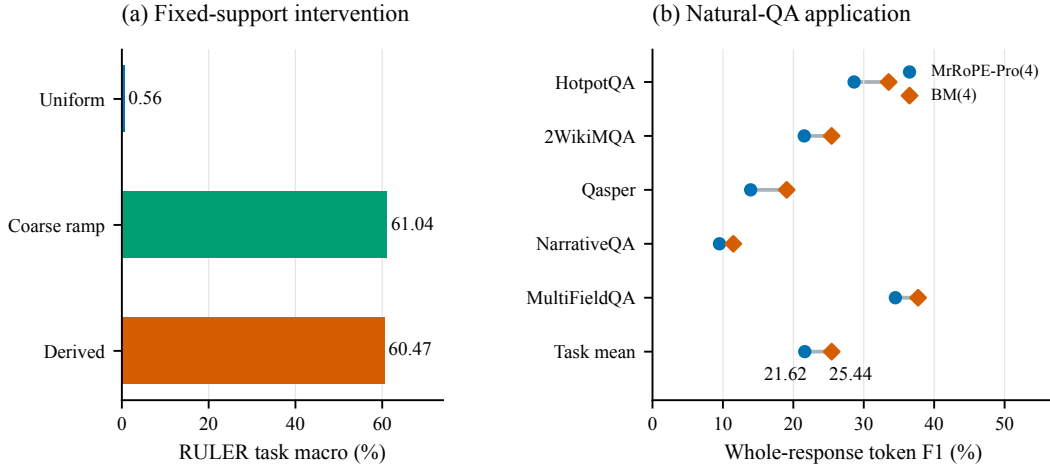


Figure 11: **Fixed-support identification and natural-QA use are separate comparisons.** (a) Frozen OLMo 16K scores on nine held-out RULER tasks (20 rows/task), with endpoints, amplitude, weights and inputs shared. (b) Matched BM/MrRoPE-Pro $s = 4$ natural QA on 631 untruncated inputs longer than 4K, using whole-response token F1. The two panels retain their own task and metric definitions.

Table 31: Qwen-0.5B RULER-13 per-task score (%), 20 examples per cell.

Task	Native	32K		Native	64K	
		Index	YaRN		Index	YaRN
Single-1	100.00	100.00	100.00	75.00	100.00	100.00
Single-2	100.00	100.00	100.00	15.00	100.00	75.00
Single-3	100.00	100.00	100.00	30.00	100.00	95.00
MultiKey-1	90.00	95.00	90.00	50.00	90.00	65.00
MultiKey-2	50.00	60.00	35.00	0.00	40.00	25.00
MultiKey-3	5.00	0.00	10.00	0.00	5.00	5.00
MultiValue	62.50	81.25	81.25	22.50	63.75	58.75
MultiQuery	62.50	80.00	77.50	22.50	65.00	50.00
Variable tracking	48.00	42.00	46.00	3.00	28.00	36.00
Common words	2.50	2.00	2.50	7.00	0.50	0.00
Frequent words	46.67	31.67	35.00	31.67	46.67	40.00
QA-1	20.00	10.00	30.00	5.00	15.00	20.00
QA-2	25.00	25.00	20.00	25.00	15.00	20.00

G.2 PLACEMENT ON ANOTHER FINITE PAIR GRID

The alternative direct-gap construction evaluates the same clipped profile using the target table’s local frequency spacing:

$$c_K = (1 - b^{-1/K})^{-1}, \quad \xi_k^{(K)} = \log \frac{L_{\text{ref}} \omega_k^N}{2\pi c_K}, \quad m_k^{\text{gap}} = \text{clip} \left(\frac{\xi_H - \xi_k^{(K)}}{\xi_H - \xi_L}, 0, 1 \right).$$

Both placements share the frozen ξ_H, ξ_L and amplitude coefficient 0.074. Within each comparison, endpoints, scale, amplitude, checkpoint, inputs, decoder, and scorer are matched. Table 32 reports two independent-input confirmations with 80 rows per task.

Qwen-0.5B uses the 32-pair, $s = 2$ setting above and data seed 202609026, with four tasks at each of 32K and 64K. These are separate inputs from the subsequent RULER-13 confirmation. Gemma-1.1-2B-Instruct uses $K = 128$, $b = 10,000$, a fixed operating reference of 4096, $s = 4$, and $g = 1 + 0.074 \log 4 = 1.1025857827$. Its confirmation uses data seed 202609028 at 16K. The task vector is NIAH single-1, multikey-2, multikey-3, and variable tracking. Gemma’s direct-gap scores are 100/88.75/20/82.50%; normalized-index scores are 100/88.75/40/87.25%. Both tables

were frozen before these inputs were generated, and the earlier selection rows are excluded. This confirmation compares the two placements; the Native/YaRN baseline panel uses a different, earlier input set.

Table 32: Independent-input placement comparisons, 80 examples per task and four tasks per panel. Intervals are for index minus direct-gap, in percentage points: 97.5% for each Qwen length (Bonferroni over two lengths), 95% for Gemma.

Model	Length	Direct-gap (%)	Index (%)	Difference interval
Qwen-0.5B	32K	49.69	53.37	$[-0.25, +7.75]$
Qwen-0.5B	64K	46.62	46.13	$[-4.88, +3.81]$
Gemma-2B	16K	72.81	79.00	$[+2.81, +9.63]$

G.3 BOUNDARY-MATCHED INTERMEDIATE EXPONENTS

Let l be the last native frequency pair making more than 32 turns over the reference window and h the first making less than one turn. Write $N = h - l$ and $q = \text{clip}(k - l, 0, N)$. All profiles leave the fast band unchanged and divide the slow-band frequencies by s :

$$\omega'_k = \omega_k^N s^{-m_q}, \quad m_0 = 0, \quad m_N = 1. \quad (57)$$

With the same band boundaries, the three intermediate allocations are

$$m_q^{\text{Uni}} = \frac{q}{N}, \quad (58)$$

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad (59)$$

$$m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (60)$$

The Uni and Pro rules follow the uniform and progressive mixed-radix constructions of Tian et al. (2026). BM smooths the radix increments $\epsilon_q = m_q - m_{q-1}$ at both band boundaries.

To obtain BM, minimize $\sum_{q=0}^N (\epsilon_{q+1} - \epsilon_q)^2$ with $\epsilon_0 = \epsilon_{N+1} = 0$ and $\sum_{q=1}^N \epsilon_q = 1$. The discrete Euler equation gives

$$\epsilon_q = \frac{6q(N+1-q)}{N(N+1)(N+2)};$$

summing gives (60). The quadratic form is strictly convex with the endpoints fixed, so this solution is unique. Figure 12 shows the actual profile shapes for the tested transition widths.

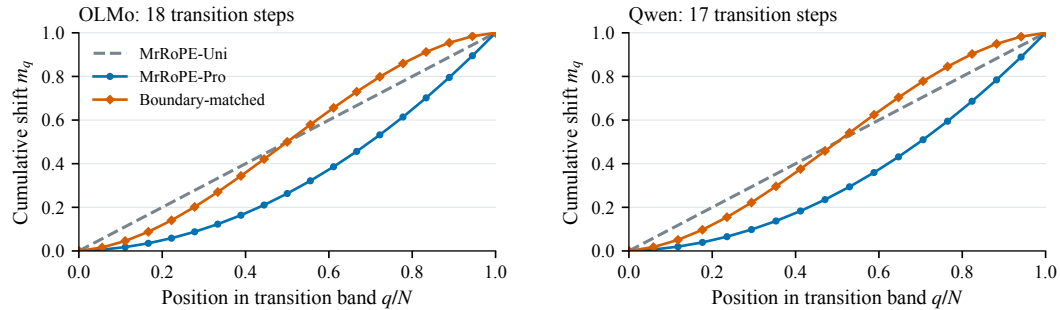


Figure 12: **Intermediate-band exponent displacements.** All three profiles have the same cumulative scale at the endpoints. BM redistributes the increments toward the middle of the transition band.

OLMo-2-0425-1B-Instruct uses $b = 500,000$, reference length 4096, $K = 64$, and $(l, h, N) = (14, 32, 18)$. Qwen-2.5-3B/7B-Instruct use $b = 10^6$, reference length 32768, $K = 64$, and $(l, h, N) = (23, 40, 17)$. The factor-four comparisons use $g = 1 + 0.1 \log 4 = 1.1386294361$ for both MrRoPE-Pro and BM, with FP32 frequency tables and BF16 model execution.

G.4 SIX-TASK MODEL COMPARISONS

The shared task types are NIAH single-2, multikey-2, multiquery, variable tracking, frequent words, and QA-1. OLMo’s confirmation uses seed 20260910 with four short and eight long examples per task, after a separate development panel with seed 20260909. Its 4K/16K scores are listed in Table 20; MrRoPE-Uni on the same confirmation inputs scores 76.88/32.12%, and official YaRN scores 54.38/6.94%. The Qwen-3B panel has two 32K and four 128K examples per task. The 7B screen reuses the first one short and two long examples per task; this selection was fixed before its outputs were inspected. Table 33 reports every task mean.

All arms use the corresponding native chat template and the same task-specific generation caps, score the entire saved response with the RULER task metric, and retain generated token IDs and EOS flags. On Qwen-7B, a tokenwise MLP chunk size of 4096 makes 128K evaluation fit in memory; both arms share that path while retaining full attention.

Table 33: All six-task BM/MrRoPE-Pro comparisons. Values are official RULER task scores (%); model-specific sample counts are given in Table 20.

Model	Task	Short		Long	
		MrPro	BM	MrPro	BM
OLMo-1B	Single-2	75.00	100.00	0.00	100.00
	MultiKey-2	25.00	100.00	0.00	37.50
	MultiQuery	18.75	87.50	0.00	43.75
	Variable tracking	25.00	70.00	0.00	10.00
	Frequent words	58.33	58.33	16.67	41.67
	QA-1	25.00	75.00	0.00	75.00
Qwen-3B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	75.00	50.00
	MultiQuery	100.00	100.00	93.75	100.00
	Variable tracking	90.00	100.00	75.00	75.00
	Frequent words	83.33	100.00	75.00	75.00
	QA-1	50.00	50.00	50.00	25.00
Qwen-7B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	100.00	100.00
	MultiQuery	100.00	100.00	100.00	100.00
	Variable tracking	100.00	80.00	90.00	60.00
	Frequent words	100.00	100.00	66.67	66.67
	QA-1	0.00	0.00	50.00	0.00

G.5 NATURAL QUESTION ANSWERING

The natural comparison reuses frozen, source-verified LongBench prompts (Bai et al., 2024). Inputs are untruncated, retain their source questions and references, and fit the 16K input-plus-generation budget. Selection follows the pre-existing source-hash order; no generated answer participates in selection. The complete eligible pool contains 778 questions, of which 631 have more than 4096 input tokens. Both methods retain the same factor-four table at every length.

Greedy generation uses a 32-token cap for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. The metric is token F1 on the complete saved generated response against the task references. Table 34 gives the long-input and original-window-length strata, including EOS counts. The long-input macro averages the five task means; the shorter-input macro averages its four available tasks. The respective BM-minus-MrPro changes are +3.8194 and +2.0028 points, with paired within-task bootstrap intervals [1.3161, 6.2920] and [−3.5618, 7.4742]. The MultiFieldQA long-input row includes 40 capped responses in each arm; these remain in the F1 calculation.

Table 34: Natural QA under static $s = 4$. Token F1 uses the whole generated response; EOS counts are recorded separately.

Input length	Task	n	MrPro F1 (%)	BM F1 (%)	MrPro EOS	BM EOS
4K–16K	HotpotQA	166	28.63	33.54	160	163
	2WikiMQA	173	21.55	25.43	169	171
	Qasper	119	13.94	19.06	119	119
	NarrativeQA	61	9.51	11.48	61	61
	MultiFieldQA	112	34.49	37.72	72	72
$\leq 4K$	HotpotQA	5	58.33	51.67	5	5
	2WikiMQA	26	34.62	44.98	26	26
	Qasper	78	26.16	28.14	78	78
	MultiFieldQA	38	50.94	53.27	35	33

G.6 COMPACT DESCRIPTIONS OF A USEFUL ALLOCATION

An ordered 64-slot movement profile can be approximated by a compact law without retaining every local feature. Define $c_k = L\omega_k/(2\pi)$, $c_{\text{orth}} = (1 - b^{-1/K})^{-1}$, and $\xi_k = \ln(c_k/c_{\text{orth}})$. The adjacent-frequency separation is exactly $L(\omega_k - \omega_{k+1})/(2\pi) = e^{\xi_k}$. A clipped-affine law fitted to the frozen OLMo movement vector is

$$G_4(\xi) = \text{clip}\left(\frac{\xi_H - \xi}{\xi_H - \xi_L}, 0, 1\right), \quad \omega'_k = \omega_k 4^{-G_4(\xi_k)}.$$

The fitted boundaries are $\xi_H = 0.7382780681$ and $\xi_L = 0.3664038351$. The two parameters were frozen before examining Qwen geometry or model results. Both checkpoints use the fixed cosine/sine gain $1 + 0.074 \ln 4$. The fit has movement MAE 0.001223 on OLMo. The original OLMo-to-Qwen transport interpolates the full profile in $u = e^\xi$; C2 evaluates the frozen fitted law directly in ξ .

Table 35: **A compact profile preserves long-context behavior.** Official task macro (%). OLMo uses RULER-13; Qwen uses four core tasks. Long results are post-gate diagnostics with the already frozen tables; Qwen 32K was collected after the long results.

Model	Endpoint	Full 64-point profile	C2
OLMo	8K	66.71	66.84
OLMo	16K	49.86	50.24
Qwen	32K	70.00	71.25
Qwen	64K	67.25	67.75
Qwen	128K	54.50	57.25

The long-end results show that detailed 64-point structure is not required to preserve these measured behaviors. At the native window, OLMo’s PPL retention is 0.870971 and task retention 0.904932; Qwen native-task macro is 82.00% versus 71.25% for C2. The registered deployment requirement was retention at least 0.875, so the single-table deployment gate failed. This operating verdict and the positive low-dimensional long-behavior result answer different questions. The original profiles, C2, and the later normalized-index placements retain distinct tensor identities.

G.7 AN INTERIOR-PROFILE CONTROL AT FIXED TOTAL DISPLACEMENT

For $K = 64$, let $k = 1, \dots, 18$, $p_k = 6k(19 - k)/(18 \cdot 19 \cdot 20)$, and $v_k = k - 19/2$. Two increment profiles are

$$\epsilon_k^A = p_k \left(1 - \frac{10}{119}v_k\right), \quad \epsilon_k^B = p_k \left(1 - \frac{10}{119}v_k + \frac{35}{1496} \left(v_k^2 - \frac{357}{20}\right)\right).$$

Both use $m_j = 0$ through slot 14, cumulative increments on slots 15–32, and $m_j = 1$ thereafter. They have the same increment mass and centroid, identical sampled endpoints, and $\sum_j m_j = 42$. Installed frequencies are $\omega'_j = \omega_j 4^{-m_j}$ with fixed gain $1 + 0.1 \ln 4$ and terminal shift $m = 1$. This is a controlled comparison within that displacement setting.

On 350 shared 16K OLMo development inputs, scores are 43.6714% and 54.4000%. The paired difference is +10.7286 points, with 75 wins, 26 losses, and 249 ties; the paired standardized mean is $t = 5.47$. A separate 16-document continuous-text evaluation reports NLL 2.9417 versus 2.8309. These observations distinguish two internal profiles that share their total displacement. They are not a held-out method-selection win: the task panel participated in profile exploration. The numerical construction, row-score aggregation, and common-total checks are reproduced by `figs/verify_recovered_assets.py`.

G.8 FOUR METHODS ON THE SAME OLMo TASK PANEL

The six-task panel contains 24 prompts at 4K and 48 at 16K. All four methods use $s = 4$ and gain $1 + 0.1 \ln 4$. MrRoPE profiles are local formula implementations; official YaRN denotes the channel-index ramp implementation. Table 36 is recomputed from the recorded per-task aggregates, with matched BM/control comparisons recorded on the shared row identities.

Table 36: **Intermediate-profile comparison on the OLMo six-task panel.** Official task macro in percent, including partial-credit tasks.

Method	4K	16K
MrRoPE-Pro(4)	37.85	2.78
Official YaRN(4)	54.38	6.94
MrRoPE-Uni(4)	76.88	32.12
BM(4)	81.81	51.32

An equal-total-displacement shape control. For $q = 1, \dots, n - 1$, summing the cumulative profiles gives

$$\sum_q m_q^{\text{BM}} = \sum_q m_q^{\text{Uni}} = \frac{n-1}{2}, \quad \sum_q m_q^{\text{Pro}} = \frac{n-1}{3}.$$

The outer bands agree, so BM and Uni have identical full-table displacement, whereas BM exceeds Pro by $(n-1) \log s/6$. On the actual $K = 64$, $[l, h] = [14, 32]$ grid, full-table $\sum_k m_k$ is 40.5 for BM/Uni and $37\frac{2}{3}$ for Pro. The 16K BM–Uni difference of 19.20 points therefore compares different shapes at equal total displacement. This is the recorded small six-task panel, separate from natural QA and the C42 development panel; it does not prove that boundary smoothness alone explains the task gain.

The official YaRN and Uni raw-output files have recorded hashes but are not bundled with the source archive; their per-task aggregates and identities are preserved. The separate natural-QA experiment in Appendix G.5 evaluates the BM/MrRoPE-Pro comparison on natural inputs.

G.9 FULL-LAG POSITIONAL GEOMETRY AS A CANDIDATE CONSTRUCTION

FullLagP2 constructs the native sine–cosine column pairs across all separations $d = 0, \dots, 32767$ with causal weights proportional to $32768 - d$. For each pair it computes its least-squares residual fraction relative to the other pairs (cutoff 10^{-10}), min–max normalizes these fractions to $\bar{u}_j$, and sets $m_j = (1 - \bar{u}_j)^2$. The resulting full table uses $s = 4$ and gain $1 + 0.074 \ln 4$. It depends on positional geometry, with no model-weight or activation fitting.

This is a positive small-panel example of converting full-pair geometry into a useful allocation. With the same full array on Qwen-3B at 64K (four prompts/task), FullLagP2/MrPro scores are 75/50% for MultiKey-2, 90/95% for variable tracking, and 66.67/75% for frequent words. The decoder uses repetition penalty 1.1; these outputs remain separate from later screening panels. The paired score fractions and checkpoint identities are preserved in `figs/recovered_asset_inputs.json`. The construction’s geometric residual is a design statistic, and the task measurements establish where it is useful.

Table 37: **Recorded FullLagP2 task outcomes on Qwen-1.5B.** Eight prompts/task/length, official task score in percent. The matched-gain MrPro arm uses the same gain as FullLagP2.

Length	Task	MrPro	MrPro, matched gain	FullLagP2
64K	MultiKey-2	12.50	25.00	37.50
	Variable tracking	82.50	77.50	87.50
	Frequent words	45.83	45.83	70.83
128K	MultiKey-2	0.00	–	0.00
	Variable tracking	72.50	–	85.00
	Frequent words	50.00	–	50.00

H TAILSPLINE: CONSTRUCTION AND FROZEN EVALUATION

H.1 FINITE-GRID PROOF

Let $D\epsilon = (\epsilon_{q+1} - \epsilon_q)_{q=1}^{n-1}$ and $H = D^\top D + e_n e_n^\top$. A zero value of $\epsilon^\top H \epsilon$ requires constant increments and a zero terminal increment, hence $\epsilon = 0$. Thus H is positive definite. For $n > 1$, the proposed $\epsilon_q = c(n+q)(n-q+1)$, with $c = 3/[n(n+1)(2n+1)]$, satisfies

$$\epsilon_1 - \epsilon_2 = 2c, \quad 2\epsilon_q - \epsilon_{q-1} - \epsilon_{q+1} = 2c, \quad 2\epsilon_n - \epsilon_{n-1} = 2c.$$

Therefore $H\epsilon = 2c\mathbf{1}$, the equality-constrained stationarity equation. The increments are positive and sum to one; summing them yields Eq. (11). The objective minimum is $2c = 6/[n(n+1)(2n+1)]$. For $n = 1$, the constraint gives $\epsilon_1 = 1$, agreeing with the formulas. Positivity makes additional nonnegativity constraints inactive.

Adding ϵ_1^2 yields the symmetric objective, with unique solution

$$\epsilon_q^{\text{BM}} = \frac{6q(n-q+1)}{n(n+1)(n+2)}, \quad m_q^{\text{BM}} = \frac{q(q+1)(3n+2-2q)}{n(n+1)(n+2)}. \quad (61)$$

Both profiles are monotone with $m_0 = 0, m_n = 1$. The proof determines the solution given a boundary prior; it does not select that prior by task accuracy. The portable CPU check independently solves the constrained tridiagonal system and compares it with each closed form.

What the objective smooths. On the geometric native grid, Eq. (9) makes each change between neighboring log gaps equal to $(\epsilon_{q+1} - \epsilon_q) \log s$. The terminal change is $-\epsilon_n \log s$ because the fully interpolated tail has constant displacement. Thus $(\log s)^2$ times Eq. (10) is exactly the squared log-gap variation within the transition and at its low-frequency junction. The high-frequency junction is deliberately unpenalized. This statement concerns spectral geometry, not a measured attention-loss mediator.

At the terminal gap,

$$\epsilon_n^{\text{Pro}} = \frac{2}{n+1}, \quad \epsilon_n^{\text{TS}} = \frac{6}{(n+1)(2n+1)}, \quad \frac{\epsilon_n^{\text{TS}}}{\epsilon_n^{\text{Pro}}} = \frac{3}{2n+1}.$$

The opposite entry behavior is also explicit: $\epsilon_1^{\text{Pro}} = 2/[n(n+1)]$, whereas $\epsilon_1^{\text{TS}} = 3/(2n+1)$. TailSpline trades a larger entry jump for a smaller terminal jump. A useful boundary prior need not minimize both. In the continuum limit, $m^{\text{TS}}(u) = (3u - u^3)/2$ solves $\min \int_0^1 (m''(u))^2 du$ under $m(0) = 0, m(1) = 1, m'(1) = 0$; the natural entry condition is $m''(0) = 0$. The finite-grid formula is used in every model experiment, not this limiting polynomial.

H.2 PROTOCOL AND COMPLETE LENGTH RESPONSES

Frozen Llama-3-8B-Instruct has native window 8192. Both arms use exact finite-grid profiles, $s = 4$, canonical 32/1-turn band [18, 35], and cosine/sine gain $1 + 0.1 \ln 4$. Each table is installed before prefill and shared across layers, requests and evaluation lengths. Construction uses no weights, activations or calibration outputs.

RULER-13 contains 10 prompts per task at 8/16/32K: 390 paired inputs per arm. NIAH and passkey are subsets of this panel. The PPL panel contains 32 ProofPile-test and 14 PG19-test documents at each length: 138 rows per arm. Both compared methods retain their $s = 4$ table even at the native input length.

Normalized trapezoidal log-length AUC weights the three endpoints by $1/4, 1/2, 1/4$. For PPL it integrates PPL over log length, not log PPL. At each length, PPL is the exponential of total whole-sequence loss divided by total scored tokens across the 46 documents. The paired document bootstrap resamples document identities jointly across lengths, recomputing token-weighted NLL, PPL and AUC for each draw. Task aggregates weight tasks equally. The recorded paired intervals characterize the tested inputs and task families for this frozen checkpoint. Family AUC differences are retrieval

Table 38: Recorded Llama responses. NIAH uses recorded paired differences; absolute scores are not reconstructed.

Endpoint	8K	16K	32K
RULER-13 MrPro (%)	87.5000	80.0897	54.7308
RULER-13 TailSpline (%)	89.7436	84.0513	57.3590
NIAH TailSpline-MrPro (pp)	-0.3125	+5.9375	+3.4375
PPL MrPro	5.286429	4.518111	4.161831
PPL TailSpline	5.295283	4.530318	4.110622

+3.750 points, tracking +6.500, aggregation approximately +3.792, and QA -1.250. Multikey-2 improves by 22.5 points; QA-1 and multikey-1 each decrease by 5.0. TailSpline/MrPro have 30/37 empty outputs, 388/384 EOS-terminated outputs and 2/6 budget-limited outputs, each out of 390. Official scores are distinct from complete-answer-plus-EOS exactness.

The portable input `figs/allocation_value_inputs.json` records reported aggregates and original source identities. Original generation and LM streams are maintained separately. The packaged frequency constructor computes the native-grid boundaries and FP32 table from public base, pair count, native length and scale; no model object is required. Regeneration checks aggregate arithmetic and algebra, not a new model execution. TailSpline and MrPro also differ in total log-frequency displacement: this comparison tests complete profiles, not boundary regularization as an isolated task mechanism.

H.3 PROSPECTIVE OLMo CONFIRMATION

The OLMo-2-0425-1B-Instruct comparison uses original window 4096, canonical band [14, 32], $s = 4$ and the same gain as Llama. Generation batch size is four. The methods and data were frozen before reading outputs from this experiment; historical BM/front-loaded work on OLMo had informed the direction. This is prospective confirmation of the exact table, not an untouched-model blind test. Its tokenized prompts are not paired with Llama’s prompts across models. Full-13 AUC

Table 39: Complete OLMo length responses under the frozen two-arm contract.

Endpoint	4K	8K	16K
RULER-13 MrPro (%)	32.5128	14.8462	7.2436
RULER-13 TailSpline (%)	72.6667	72.5513	48.6154
NIAH MrPro (%)	33.1250	13.1250	7.1875
NIAH TailSpline (%)	88.7500	87.8125	60.0000
PPL MrPro	13.795244	14.356459	12.827048
PPL TailSpline	11.152472	10.117407	8.536196

difference is 0.492340 with paired 95% interval [0.453365, 0.530449]; NIAH difference 0.644531 has interval [0.592188, 0.694531]; PPL difference -3.852932 has interval [-4.775042, -3.125459]. Passkey AUC is 0.950 versus 0.375. All 13 task differences are positive, ranging from +0.0325 for common words to +0.875 for single-3. Family differences are retrieval +0.64453, tracking +0.37000, aggregation approximately +0.16208, and QA approximately +0.27563.

Neither arm has empty outputs. TailSpline/MrPro terminate with EOS on 257/248 of 390 inputs and hit the cap on 133/142. The PPL data use the same document sources as Llama but OLMo-tokenized prefixes. Total transition displacement also differs between the profiles, so the large observed advantage does not isolate the roughness mechanism. Report and original stream identities accompany the portable aggregates.

H.4 EVALUATION IMPLEMENTATION

Generation is greedy, with repetition penalty 1, no repeat-ngram restriction and the model’s EOS IDs. Llama uses batch size one with cached 8192-token prefill chunks; OLMo uses batches of at most four identical-length inputs. The static FP32 frequency table is installed after model loading and its installed values are checked against the receipt. NIAH point-depth tasks use two prompts at each target depth 10/30/50/70/90% per length. Other tasks keep their recorded task-specific output budgets.

Full-13 intervals use 20,000 paired resamples within each task-length cell, retaining fixed task weights; NIAH intervals aggregate the corresponding retrieval tasks. The report treats different length cells separately for this panel. PPL intervals instead resample the 46 document identities jointly across all lengths, also using 20,000 draws. The two uncertainty calculations therefore preserve different experimental units. These are conditional evaluation intervals, not uncertainty across training seeds or model families.

I FREQUENCY SPECTRA AND LEARNED ASSIGNMENTS ARE DISTINCT

A positional basis enters attention through learned Q/K projections. We test three separate changes: retargeting its range, swapping tables across trained weights, and reassigning the same frequencies to different slots. Each intervention addresses a different explanation of model behavior.

I.1 CHANGING THE EVALUATION RANGE CHANGES THE PREFERENCE

Return to the paired models of §3.1, without further training. The retained-range condition uses base $b = 256$ at every evaluation length. The retargeted condition replaces it with $b = L_{\text{eval}}$ for both allocations, keeping each trained weight set and normalized shape z fixed (the exact rule is in Appendix B.1). Cosh-minus-geometric NLL becomes $+0.060/+0.227/+0.460$ at 512/1024/2048 tokens: all three seeds now favor the geometric allocation (Fig. 13a). The same trained pair produces opposite orderings under the two range policies. Allocation and range are separately controllable, but their effects interact in the learned model.

I.2 CROSSING WEIGHTS WITH THEIR FREQUENCY TABLES

To isolate learned compatibility, hold each weight set fixed and cross the runtime tables. In a 50M pair, Geo and Cosh achieve matched PPLs of 7.14 and 7.16. Installing the Cosh table into Geo-trained weights raises PPL to 76.20; the reverse swap gives 23.05. A 151.9M crossing repeats the preference across two training seeds using factor-four tables derived from each training allocation (Fig. 13b). The two seed-specific interaction contrasts are 3.400 and 3.251 tail NLL. These crossed comparisons expose the learned association: both allocations support competent models, and each model learns to use its own basis (full crossings in Fig. 9).

I.3 FREQUENCY IDENTITY AND SLOT IDENTITY

The distinction becomes sharper when the frequency multiset itself is preserved. For a block permutation P corresponding to a frequency permutation π ,

$$\mathcal{R}_{\pi\Omega}(\Delta) = P\mathcal{R}_{\Omega}(\Delta)P^{\top}, \quad (Pq)^{\top}\mathcal{R}_{\pi\Omega}(\Delta)(Pk) = q^{\top}\mathcal{R}_{\Omega}(\Delta)k. \quad (62)$$

Thus permuting Q/K blocks together with frequencies exactly preserves the attention kernel. With weights fixed, however, permuting only internal frequency assignments raises OLMo tail NLL from 3.10423 to 6.86493 and lowers Qwen 64K task macro from 0.7000 to 0.0000. The model uses a specific association between coordinate subspaces and frequencies (Fig. 13c,d).

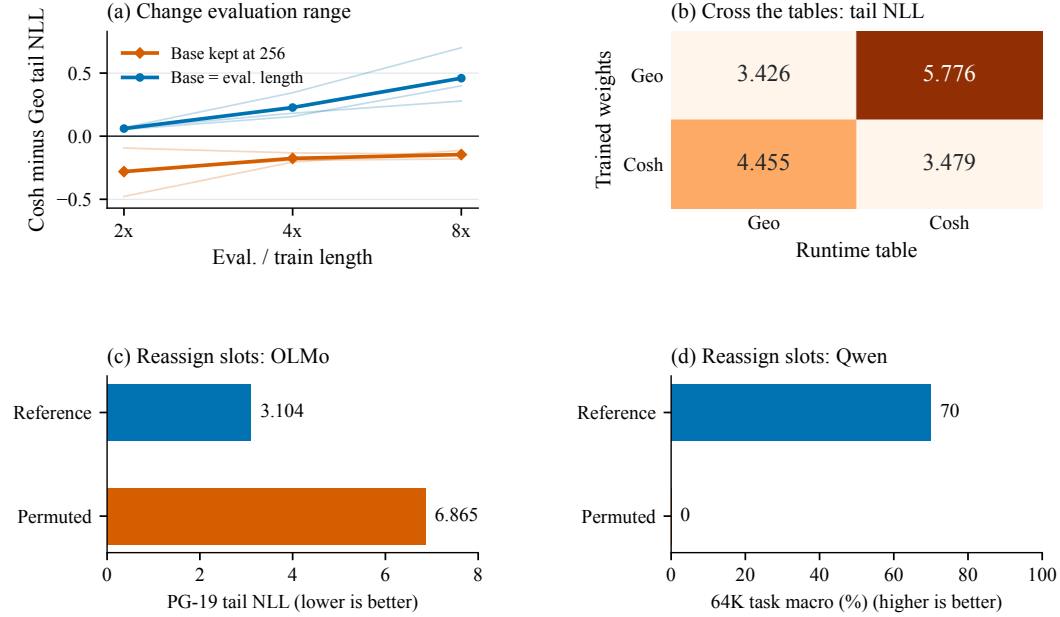


Figure 13: **Three interventions distinguish the supplied spectrum from learned compatibility.** (a) Range retargeting reverses the ordering of the same trained allocations; thin lines are three seeds and thick lines their means. (b) Crossed factor-four tables at 1K: rows are trained weights, columns runtime tables, and cells are mean tail NLL over two seeds and 32 anchors/seed. (c,d) Permuting interior frequency assignments preserves the final frequency multiset, endpoints and gain, but changes OLMo PG-19 tail NLL and Qwen 64K task macro (4 tasks, 20 rows/task). Each metric has its own axis. Joint Q/K compensation is the algebraic identity in (62), not a plotted recovery experiment.

For genuinely different frequency multisets, position-independent invertible Q/K maps cannot equate the rotation kernels over an interval containing zero (Theorem 5). Allocation changes the supplied basis; assignment changes its association with learned coordinates. The experimental counterparts establish why a design can be valuable during learning yet require a different installation strategy on a mature checkpoint. These controls complement the constructive learning and deployment results in the main text.

J FIXED-TABLE QUALITY OVER A DEPLOYMENT INTERVAL

J.1 THE DEPLOYMENT OBJECTIVE AND ITS DESIGN VARIABLES

Let L be a checkpoint’s native window and $H = SL$, $S > 1$, its declared deployment horizon. For a task-valid minimum length $\ell_{\min} > 0$, the objective concerns $[\ell_{\min}, H]$. The same table and amplitude are used at every length; neither changes with the current prompt length or during decoding. For a fixed checkpoint, write

$$\omega'_k = \omega_k^N S^{-m_k}, \quad m_k = cb_k, \quad 0 \leq b_0 \leq \dots \leq b_{K-1} = 1, \quad 0 \leq c \leq 1. \quad (63)$$

Within this monotone family, active increments of b locate the transition, their normalized distribution specifies its shape, and c controls the terminal compression. This is a search parameterization; it does not assert that an optimal unrestricted table must have one contiguous band. Changing c scales both the transition and the slow plateau. Comparisons of depth therefore measure whole-profile depth effects, rather than an isolated low-frequency intervention. Allowing the terminal depth to vary also changes sampled support, unlike the identification intervention in §3.1.

For higher-is-better task quality $Q_\theta(r)$ at length rL , one useful interval summary is

$$A_{[1,S]}(\theta) = \frac{1}{\log S} \int_1^S Q_\theta(r) d \log r. \quad (64)$$

This extended-window summary accompanies evaluations below L , native and endpoint values, the worst measured length, and task-family changes. Different tasks can have different valid short-input ranges. We do not extrapolate an AUC to one-token prompts or combine NLL and task accuracy into a single score. On a finite length grid, we use trapezoidal integration in log length. The measured minimum is not a certificate of the continuous interval minimum.

Baseline curves use a fixed table under the same evaluation contract. A pointwise envelope of several baselines is an oracle reference for regret, not a deployable method that may switch tables. For RULER we score the complete generated response with the official task scorer; normalized full-answer exactness, EOS, cap exhaustion, and natural-QA response F1 are separate measurements. This distinction matters when a model produces the requested answer and then continues generating.

J.2 A CONDITIONAL ALTERNATIVE TO FULL TERMINAL COMPRESSION

A slow plateau with frequency multiplier $a = S^{-c}$ maps separation rL to native effective separation arL . Full compression $a = 1/S$ exactly matches native phase exposure at the endpoint. It need not minimize distortion over the intervening lengths.

Proposition 11 (Endpoint and interval optima for a scalar slow-block objective). *Let $S > 1$, $a \in [1/S, 1]$, and let F be continuously differentiable on $[1/S - 1, S - 1]$. Define*

$$J(a) = \frac{1}{\log S} \int_1^S F(ar - 1) \frac{dr}{r}. \quad (65)$$

Then

$$J'(a) = \frac{F(aS - 1) - F(a - 1)}{a \log S}. \quad (66)$$

If $F(z) = f(|z|)$ with f strictly increasing, the unique minimizer is

$$a_* = \frac{2}{S + 1}, \quad c_* = \log_S \frac{S + 1}{2}. \quad (67)$$

The endpoint objective $F(aS - 1)$ is instead uniquely minimized at $a = 1/S$.

Proof. Differentiating under the integral and substituting $z = ar - 1$ gives $J'(a) = (\log S)^{-1} \int_1^S F'(ar - 1) dr$, hence (66). On the feasible interval, $a - 1 \leq 0 \leq aS - 1$. Strict increase of f makes the derivative negative when $aS - 1 < 1 - a$ and positive when the inequality reverses. Equality gives a_* and establishes uniqueness. At the endpoint, zero error is attained uniquely at $a = 1/S$. $\square$

This balances pull-in at the native length against overshoot at the horizon in a declared scalar objective. It does not identify the true task loss of a checkpoint. For $S = 2, 4, 8$, the conditional depths are 0.584963, 0.660964, and 0.723308, respectively. These values define concrete experimental controls against $c = 1$; they are not tuned task results.

Native and endpoint emphasis change the answer. For any probability measure over $r \in [1, S]$, a quadratic objective with native and endpoint weights $\lambda, \eta \geq 0$ has

$$J_{\lambda, \eta}(a) = \mathbb{E}[(ar - 1)^2] + \lambda(a - 1)^2 + \eta(aS - 1)^2, \quad (68)$$

$$a_* = \frac{\mathbb{E}[r] + \lambda + \eta S}{\mathbb{E}[r^2] + \lambda + \eta S^2}. \quad (69)$$

The second derivative is strictly positive, and $a_* \in [1/S, 1]$. Under log-uniform measure and $\lambda = \eta = 0$, this reduces to (67). Native emphasis moves the optimum toward 1; endpoint emphasis moves it toward $1/S$. Thus the chosen operating distribution and preservation priorities belong in the objective, rather than being determined by S alone. This scalar analysis concerns exposure relative to L ; real short-input task quality still requires direct measurement.

An asymmetric example is $F_\kappa(z) = \kappa z^2$ for $z \geq 0$ and z^2 otherwise, with $\kappa > 0$. Equation (66) yields

$$a_* = \frac{1 + \sqrt{\kappa}}{1 + S\sqrt{\kappa}}.$$

Table 40: **A fixed table’s endpoint and interval quality can differ.** Official task scores in percent; each row uses one table at every listed length. Llama is a small development screen (12 prompts/length); OLMo is a development mini panel (108 prompts/length). Within each panel the inputs, scorer, and decoder are shared. All remapped profiles use the registered C42V24 transition shape. AUC is integrated over the listed log-length grid, using unrounded cell means in the original summaries. These panels are distinct from the earlier 350-row and natural-QA studies.

Llama, fixed $S = 8$	8K	16K	32K	48K	64K	AUC
C42V24 [14, 32]	95.83	97.92	100.00	95.83	81.25	96.62
C42V24 [16, 34]	97.92	97.92	87.50	95.83	81.25	93.67
OLMo, fixed $S = 4$	4K	8K	16K	AUC		
BM	81.67	71.31	48.50	68.20		
MrRoPE-Pro	42.28	25.63	8.18	25.43		
C42V24 [14, 31]	79.65	71.84	53.15	69.12		

A larger overshoot penalty moves the optimum toward full compression. The formula does not specify a checkpoint’s κ ; that quantity requires an independently justified measurement or loss model.

A circular counterexample. Even symmetry alone is insufficient. Set $F(z) = 1 - \cos(2\pi z)$ and $S = 4$. At $a = 2/5$, the two endpoint errors are $\pm 3/5$, so $J'(a) = 0$, but

$$J''(2/5) = \frac{25\pi \sin(6\pi/5)}{\log 4} < 0.$$

The balanced point is a strict local maximum. Consequently, neither periodicity nor successful endpoint phase matching implies the monotone distortion assumed by Proposition 11. The proposition supplies an interpretable comparison, while finite-phase and task measurements decide whether it helps a real model.

J.3 RECORDED DEVELOPMENT EVIDENCE FOR INTERVAL EVALUATION

Table 40 records two separate development panels. The Llama-3-8B-Instruct $S = 8$ comparison holds the C42V24 transition shape and gain fixed while remapping its band. The tasks are single-key (single-1), multi-key (multikey-1), and multi-value retrieval, with four prompts per task and length. Inference uses BF16 and greedy decoding, with chunked prefill and a full KV cache for long inputs. Both profiles reach 81.25% at 64K, but their 32K scores differ by 12.5 points. Endpoint evaluation alone misses this difference. Moving the band also changes total displacement; this is a band-remapped profile comparison, not a fixed-displacement intervention.

The OLMo-2-1.485B-Instruct $S = 4$ mini panel comprises six tasks, three lengths, and 18 prompts per cell: 324 responses per table. The tasks are single-key retrieval (single-2), multi-key retrieval (multikey-2), multi-query retrieval, variable tracking, frequent words, and QA-1. Cell means are averaged equally across tasks before computing AUC. The remapped C42V24 profile improves endpoint quality relative to BM, with a native cost. The AUC difference is +0.92 points, with task-stratified paired-row bootstrap interval $[-2.48, 4.36]$ (20,000 resamples). Task-AUC differences include -8.61 points on variable tracking and -1.39 on QA. This is a measured trade-off, not a confirmed improvement over BM across the interval.

These recorded summaries establish a reason to measure intermediate lengths and task families. They do not establish a universal band, a validated soft-depth method, or a complete causal explanation of generation. The large effect of changing a band in one checkpoint is also not evidence that the same slot indices transfer to another checkpoint.

J.4 FREQUENCY INTERACTIONS IN ATTENTION

For fixed pre-rotation Q/K coefficients, a rotary score can be written

$$z(\Delta; m) = \operatorname{Re} \sum_k C_k e^{i\Delta\omega_k^N S^{-m_k}}, \quad \frac{\partial z}{\partial m_k} = \log S \Delta\omega'_k \operatorname{Im}(C_k e^{i\Delta\omega'_k}).$$

The sign depends on content and separation. For a full row of visible keys, let J be the logit Jacobian with respect to m , including the actual attention scale and rotary amplitude. At the reference distribution p , the local attention-KL curvature is

$$H = J^\top (\text{diag}(p) - pp^\top) J.$$

It couples frequencies through key competition even though each score is a sum over pairs. If $m = T\epsilon$ is the cumulative-increment parameterization, the corresponding curvature is $T^\top HT$. Diagonal per-slot scores do not retain these interactions.

This calculation identifies a structural source of interaction: softmax normalization couples frequency changes through all competing keys. It explains why per-frequency geometric changes can combine differently in attention. Here the content coefficients remain symbolic; the Cosh and BM constructions use declared geometry and design parameters to form their tables. The fixed-support and crossing experiments test the resulting model behavior under their respective protocols.

J.5 EARLIER PHASE GEOMETRY AND PROFILE DIAGNOSTICS

For $s > 1$, $\omega'_j = \omega_j s^{-m_j}$ maps separation Δ to native effective separation Δs^{-m_j} . At $\Delta = sL_{\text{train}}$, containment within the unwrapped training interval requires $m_j \geq 1$. This is a per-pair geometric condition: distinct m_j generally map to distinct native separations. The phase derivative at $m_j = 0$ is $-\Delta\omega_j \ln s$, while the finite drift at the native table is zero.

For twelve OLMo-1B profiles on 350 shared 16K RULER inputs, a count of slots whose test-distance turns lie in $[0.25, 16]$ separates three response groups. Appendix J.6 defines this exploratory count and its reconstruction provenance. It leaves within-group rankings unresolved; it is not used to certify interval quality.

J.6 RECONSTRUCTING THE TWELVE-PROFILE RESPONSE ANALYSIS

Statistic and evaluation unit. For OLMo-2-0425-1B-Instruct, $K = 64$, $b = 500,000$, native window 4096, and test distance $D = 16384$, define

$$T_j(D; m) = \frac{D}{2\pi} b^{-j/K} 4^{-m_j}, \quad N(m) = \sum_{j=0}^{63} \mathbf{1}\{0.25 \leq T_j(D; m) \leq 16\}. \quad (70)$$

This is a band-count statistic. The thresholds are exploratory choices examined on the same twelve profiles, not calibrated-readout boundaries. All arms use $g = 1 + 0.1 \ln 4 = 1.1386294$, 350 shared row identities, seven equally sized RULER task groups, and 16K inputs. The score is the mean saved task score in $[0, 1]$, including partial credit; EOS is recorded separately. The 350 inputs estimate each profile’s task score, whereas the configuration-level analysis has twelve profiles. MrPro and BM use archived endpoint outputs; the other ten arms have separate saved outputs.

Fit and scope. Unweighted least squares of the logit-transformed mean score on N gives $\text{logit}(\bar{r}) = -20.45 + 1.19N$. Among 66 profile pairs, 39 have different counts and concordant scores, 0 have different counts and discordant scores, and 27 tie in count. The count therefore describes between-group ordering and does not resolve within-group differences. This is a descriptive fit to fractional scores, not a binomial success model or an independent predictive test. Threshold choices and coefficients were analyzed using these same profiles.

Reconstruction and provenance. The accompanying `figs/profile_diagnostic_inputs.json` records every m_j , full reconstruction digest, row-output digest, score sum, and fit coefficient. The common row-identity digest is `14736f284ac30c0c`. Matching row identities checks panel alignment in the saved outputs; the available records do not supply a new prompt-byte comparison for the archived endpoints. The analysis therefore reports the response of the recorded profiles.

Testing the interpretation beyond the fitted family. A subsequent matched continuous-text evaluation at 16K on 16 documents increased the count from $N = 17$ for BM to 22–26 by releasing slow-end interpolation. BM had NLL 2.8627; the five modified tables gave 2.8971, 2.8674, 2.8697,

Table 41: **All twelve profiles used for exploratory response fitting.** Scores are recomputed from saved row-level outputs. SHA prefixes identify reconstructed little-endian float32 frequency tensors; these are reconstruction digests, not recovered runtime tensor attestations. All twelve profiles participate in the fit.

Profile	N	Score sum	Mean (%)	EOS / 350	SHA-256 prefix
MrPro	15	24.80	7.09	197	505d1fb2dce3
beta_b0p25	16	50.65	14.47	181	a8313cac3ee8
turns_a0p5b32	16	56.15	16.04	169	25721495d712
beta_b0p5	16	81.20	23.20	150	4b0b97569a3a
turns_a1b16	16	92.80	26.51	167	a24e3329fc49
knife_both	17	134.60	38.46	173	4c9d6fa7684b
knife_taper	17	137.80	39.37	146	a3b8138d159e
knife_leak	17	138.90	39.69	120	8f92ffecc376
MrProBM	17	145.85	41.67	119	0ed902cba69b
beta_b2	17	175.05	50.01	119	1188a7d3c34c
turns_a2b32	17	178.50	51.00	129	cbd8c451c0bb
turns_a1b64	17	188.45	53.84	130	7aedd80ef093

2.8878, and 3.0372. Every modification increased the count while worsening this likelihood endpoint. The task-score fit above and this independent likelihood instrument measure different outcomes; together they show why the geometric count alone is insufficient to select a deployed table. Positional coverage remains a computable description, and its usefulness must be established through the learned model and the target task.

The archived BM endpoint is also not an exact text-replay target: a later same-name rerun differed on 24/350 row scores and 210/350 generated texts despite tiny reconstructed tensor differences. Table 41 reports the saved archived outputs; it does not pool them with reruns.